Final Report

Smart Learning Management System

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June 2026

Declaration of the Candidate & Supervisor

I declare that this report is my own work and has not been submitted in any form for another degree or diploma at any university or other institution of higher learning. To the best of my knowledge and belief, this report does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

I further declare that all sources of information used in this study have been properly acknowledged and referenced in accordance with accepted academic practices.

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To the best of my knowledge, the work presented in this report is the candidate's own work and is suitable for submission for examination.

Name of Supervisor: Mr. Asanka Dilruk

Signature: ____  Date: _____25/06/2026_____

Acknowledgment

I would like to take a moment to sincerely express my deep gratitude to all the individuals and organizations that played a vital role in the successful completion of my final year project, titled Smart Learning Management System (LMS).

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Abstract

The fast growth of information and communication technology has changed how schools and colleges handle teaching, learning, and managing academic activities. Learning Management Systems (LMS) are now important tools that help support digital education by offering one place for sharing content, communicating, assessing students, and keeping track of academic records. But A TOO Z College does not have a dedicated LMS, which causes problems in sharing learning materials, managing assignments, tracking attendance, keeping academic records, and communicating between teachers and students. These issues cause inefficiencies in the college's academic operations and lower the effectiveness of both regular classroom teaching and online learning. This project is about designing and building a Smart Learning Management System (LMS) for A TOO Z College.

The system is a web-based application that brings all academic management functions together for administrators, teachers, and students. It allows administrators to handle users, departments, subjects, courses, enrollments, announcements, and academic reports. Teachers can upload study materials, create assignments and quizzes, take attendance, and check student performance. Students can access course materials, submit assignments, take quizzes, see their attendance records, and get updates about academic announcements.

The system was built using HTML, CSS, Bootstrap 5, and JavaScript for the user interface, PHP for the server side, and MySQL as the database. A responsive design was used so the system works well on desktops, tablets, and mobile phones. The project followed a clear software development process that included analyzing requirements, designing the system, building it, testing it, and evaluating it. Several design documents were created, like use case diagrams, activity diagrams, entity-relationship diagrams, database schemas, and component-based architecture models.

These helped guide the development. Functional tests were done on main parts of the system, such as user sign-in, course management, attendance tracking, assignment

submission, quizzes, and reports. The tests showed that the system worked well and met all its functional and non-functional goals.

The Smart LMS offers a good solution to improve academic management, support communication between teachers and students, reduce the need for manual work, and help create a better digital learning environment. This project shows how using web-based technologies can help solve real educational problems and provides a good base for future improvements like mobile apps, advanced data analysis, and smart learning tools.

Keywords: Learning Management System (LMS), E-Learning, Academic Management, Web-Based Application, PHP, MySQL, Digital Education, Student Management, Assignment Management, Attendance Tracking.

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CHAPTER 01

1. Introduction

Education has changed a lot because of the fast progress in information and communication technologies. Today, schools and colleges use digital tools more than ever to help with teaching, learning, communication, and managing school activities. The growing use of online learning and mixed learning methods has shown the need for good platforms that can handle all the academic tasks in a single, organized place.

A Learning Management System (LMS) is a type of software that helps manage, deliver, and track educational content and school activities. It lets school leaders, teachers, and students work together in one place. With an LMS, you can manage learning materials, assignments, tests, attendance, updates, and reports in a clear and efficient way. These systems are now important for schools that want to make learning easier, improve communication, and support flexible learning. A TOO Z College currently uses a lot of old methods and separate digital tools for its academic work.

These methods make it hard to keep track of learning materials, watch student progress, maintain attendance records, and communicate effectively. Without a central platform for managing academic tasks, the school has trouble working efficiently and keeping up with modern education practices. To fix these problems, this project plans to design and build a Smart Learning Management System (LMS) that fits the needs of A TOO Z College. The new system will be a secure, web-based platform that brings all academic management functions together in one place. It will support different roles for administrators, teachers, and students. It will have features like course management, sharing study materials, submitting assignments, managing quizzes, tracking attendance, sending announcements, messaging, and monitoring student performance. The Smart LMS is built using up-to-date web technologies such as HTML, CSS, Bootstrap 5, JavaScript, PHP, and MySQL. The system is designed

to work well on different devices so users can access their academic information anytime and anywhere. With this solution, A TOO Z College can improve how it runs its academic activities, make communication better, cut down on manual work, and build a strong base for digital learning and future technology growth. This chapter explains the background of the study, describes the challenges the college is facing, explains why this project is important, outlines the main goals and objectives, and wraps up with a summary of what's covered.

1.1. Background

The quick progress in Information and Communication Technology (ICT) has greatly changed how education works around the world. Schools and colleges are now using digital tools more often to make teaching, learning, sharing information, and handling school work better.

One of the most useful digital tools in education is a Learning Management System (LMS). An LMS is a main place where schools can manage their academic tasks, and help teachers and students work together more easily. In the past, many schools used old ways to handle academic work like giving out materials, taking attendance, doing tests, checking student progress, and sharing school news.

These methods were okay for small schools, but they become hard to manage as more students and classes are added. Also, the need for online learning, digital classrooms, and blended learning has shown how limited traditional methods can be. Learning Management Systems were created to solve these problems by offering a complete digital space that brings all academic tasks into one place.

These modern LMS platforms let teachers share materials, create assignments and quizzes, check on student progress, take attendance, and talk to learners. Students can easily access study materials, take online tests, see their grades, and get important updates quickly. School leaders can manage users, subjects, departments, and reports in one place, while keeping control of all school data. Even though many schools now use LMS systems, A TOO Z College does not have its own LMS. Instead, the college

uses old methods and different digital tools to handle teaching materials, attendance, assignments, tests, and communication. This piecemeal way of doing things causes problems like slow work, heavy workload for staff, trouble tracking student progress, and late sharing of important info. Not having a single digital platform makes it hard for teachers, students, and school leaders. Teachers have trouble sharing materials and checking on student progress. Students find it hard to access resources, submit work, and get updates. School leaders spend a lot of time managing records and making reports using separate tools or by hand.

To fix these issues, this project suggests making a Smart Learning Management System (LMS) that fits the needs of A TOO Z College.



Figure 1 Growth of the Global E-Learning Market (Source: Global Market Insights / Statista)

Source: https://www. statista. com/topics/3115/e-learning-and-digital-education/?utm

This system will be a web-based platform that supports managing classes, delivering content, doing assessments, tracking attendance, communicating, and checking student performance. By putting all these functions in one system, the college can work better, improve communication, support digital learning, and create a better learning environment for everyone involved. The Smart LMS is built using modern

web tools like HTML, CSS, Bootstrap 5, JavaScript, PHP, and MySQL. The design is responsive, so it works well on computers, tablets, and phones. With this system, ATOO Z College can start building a strong digital learning foundation and improve how academic work and learning experiences are handled for all students and staff.

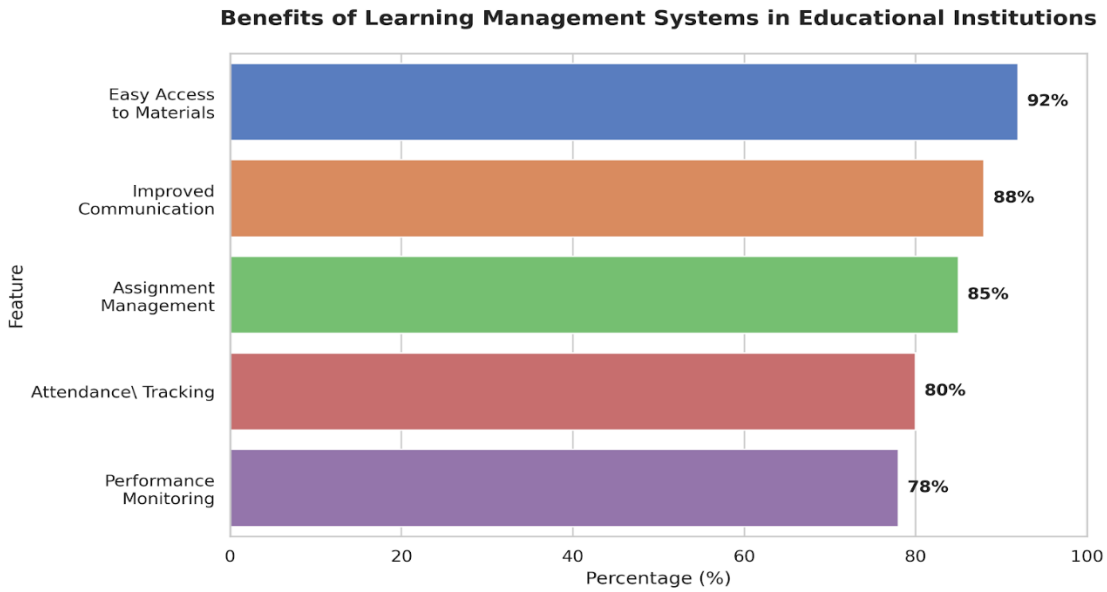


Figure 2Growth of the Global E-Learning Market (Source: Global Market Insights / Statista)

1.2. Problem Statement

Using digital technologies in education has become very important for making teaching, learning, and managing school activities better. Schools are supposed to have good systems that help in sharing knowledge, managing tests, communicating, and keeping track of student records. But many schools still use old and separate ways of doing things, which makes everything slower and harder to manage. Right now, A TOO Z College doesn't have a single system to handle all its teaching and learning activities.

Tasks like giving out study materials, keeping track of who is present, handling homework, doing quizzes, watching how students are doing, and sending important messages are done by hand or using different digital tools.

This lack of a single system causes many problems for teachers, students, and school staff. Teachers find it hard to share lessons, check in on homework, follow student progress, and record attendance properly. Using multiple tools for each task takes more time and makes their work less efficient. Also, without a special place to share updates and talk to students, communication becomes tricky and not reliable. Students face issues in getting learning materials, handing in homework, taking tests, and getting updates about their studies. Without a unified system, it's tough for them to check their progress and keep track of their learning.

These problems get worse during online or mixed learning, where having access to digital resources is very important. School staff also struggle because they don’t have one system to manage student records, course info, attendance, and performance. Doing things manually or using different apps takes a lot of time and makes mistakes more likely. Creating reports and keeping academic records properly is hard and needs a lot of effort. The main issue is that A TOO Z College doesn’t have a single, secure, and connected platform to manage all its academic and administrative needs.
The current setup doesn’t help with good communication, keeping track of student progress, managing resources, or making smart decisions based on data.

To fix these problems, this project suggests creating a Smart Learning Management System (LMS). This system will be a single web-based platform that covers all academic tasks like managing courses, sharing study materials, handling assignments, conducting quizzes, tracking attendance, monitoring student performance, sending announcements, and allowing messaging. By introducing this system, the college can make its operations faster, communication better, workload lighter, and the learning environment more effective and tech-friendly.

1.3. Motivation and Significance of Project

Motivation of the Project

The reason for creating the Smart Learning Management System (LMS) comes from the growing need for digital change in education and the issues faced at A TOO Z College. As schools and colleges use technology to improve teaching, learning, and management, there is a need for systems that can manage academic tasks well and give easy access to resources for everyone involved. Right now, A TOO Z College uses a lot of manual methods and separate digital tools to manage its academic work. This makes managing study materials, assignments, attendance, reports, and communication between teachers and students a challenge. These problems make the administrative work harder and lower the quality and efficiency of the school's services. Without a central platform, teamwork is hard, information sharing is delayed, and it's tough to track student progress. From discussions with people who need the system, it's clear that a web-based system that brings all the key academic tasks into one place is needed.

The goal of this project is to offer a simple, reliable, and easy-to-use solution that makes academic management easier and supports modern teaching and learning. Using web technologies like HTML, CSS, Bootstrap 5, JavaScript, PHP, and MySQL, the system will be affordable, easy to maintain, and adaptable as the college's needs change. This project not only solves a real problem for the institution but also helps apply the knowledge learned during the Bachelor of Information Technology program. It allows the use of software engineering, database design, web development, system analysis, user interface design, and testing in a full software project. This helps improve the institution and supports the researcher's professional growth.

Significance of the Project

The Smart Learning Management System is designed to bring many benefits to A TOO Z College by making academic management more modern and improving the quality of education. With a centralized digital platform, managing academic information becomes more efficient, reducing reliance on manual work. This also

improves the accuracy, accessibility, and security of the school's data. For administrators, the system makes managing users, departments, courses, enrollment, reports, and announcements easier. These automated features cut down on repetitive tasks and help make better, faster decisions by providing accurate academic data. Teachers get an integrated system where they can upload materials, create assignments and quizzes, record attendance, track student performance, and communicate with students.

Having all these tools in one place improves how teaching is managed and lets teachers focus more on teaching and less on administrative work. Students can access course materials, assignments, quizzes, attendance records, grades, and school announcements through a secure online environment. The system is designed to work on desktops, laptops, tablets, and smartphones, supporting both in-class and online learning. From the school's point of view, the Smart LMS supports digital transformation by offering a scalable platform for academic management.

It can grow with future needs like mobile apps, better reporting, cloud-based services, and AI-driven learning support. This project helps solve the current challenges A TOO Z College faces and provides a strong foundation for future academic development and innovation.

1.4. Aim and Objectives

1.4.1. Aim

The main goal of this project is to create, build, and put into use a Smart Learning Management System (LMS) for A TOO Z College. This system will help manage academic tasks in one place, improve communication between school staff, teachers, and students, automate important academic activities, and support modern digital learning using a secure, easy-to-use, and web-based platform.

1.4.2. Objectives

The goals of this project are:

- To look at how academic tasks are currently handled at A TOO Z College and find out the problems with the existing manual and separate systems.
- To design a web-based Learning Management System that offers secure access based on roles for administrators, teachers, and students.
- To build modules for managing users, departments, subjects, courses, student registrations, sharing study materials, handling assignments, creating quizzes, recording attendance, sending announcements, messaging, and generating academic reports.
- To give teachers tools that let them upload learning materials, create assignments and quizzes, track attendance, check student progress, and communicate well with students.
- To let students access course materials, submit assignments, take online quizzes, check attendance, view their academic records, and get updates about academic news through a single digital platform.
- To make sure the system has a design that works well on desktops, laptops, tablets, and mobile devices.
- To keep academic information safe, accurate, and dependable by using role-based login, secure data handling, and controlled access to system features.
- To test the developed Smart Learning Management System to make sure it works as expected and meets all the required features and standards, helping manage academic activities at A TOO Z College effectively.

1.5. Summary

This chapter introduced the Smart Learning Management System (LMS) project and set the stage for the rest of the report. It talked about the background of the study by showing how important Information and Communication Technology (ICT) and Learning Management Systems are in today's education. The chapter also looked at how A TOO Z College currently manages its academic activities and found out the problems with using old, manual methods and separate digital tools. The problem statement explained the shortcomings of the current system and stressed the need for a single, web-based solution to make teaching, learning, communication, and administrative work more efficient.

The motivation and importance of the project were explained by showing what benefits it could bring to administrators, teachers, students, and the college itself. The chapter also clearly stated the goals and purposes of the project to guide the design, building, use, and testing of the proposed Smart Learning Management System. In short, this chapter showed why it's important to develop the Smart Learning Management System to help A TOO Z College go digital in its academic management. The system is meant to improve how educational services are delivered, make communication better, make academic tasks easier, and offer a flexible platform that can keep up with future technology changes. The next chapter will cover the Literature Review, which will look closely at existing Learning Management Systems, related technologies, past research, and current teaching methods. This review will give the theory and real-world support needed to create and build the Smart Learning Management System.

CHAPTER 02

2. Literature Review

1.Introduction

Learning Management Systems (LMS) have become a key part of modern education because they offer digital platforms that help with teaching, learning, testing, communication, and managing academic activities. As Information and Communication Technology (ICT) keeps growing, schools and universities around the world are using LMS solutions to make education more accessible, collaborative, and of higher quality. As educational environments move towards more digital tools, Learning Management Systems have changed from basic tools for managing content into full-scale platforms that handle many aspects of education all in one system. A literature review is a detailed look at existing research, technologies, software systems, and academic papers related to a particular topic. It helps researchers understand what is already known, see what solutions exist, find their strengths and weaknesses, and discover areas where more research is needed.

Doing a thorough literature review also gives both theory and practical knowledge that helps guide the design, building, and testing of a new system. The main goal of this literature review is to look at existing Learning Management Systems and other educational technologies that are relevant to the proposed Smart Learning Management System for A TOO Z College. Several commonly used LMS platforms are examined to understand their features, what they can do, how easy they are to use, how they are set up, and their limitations. This analysis gives useful ideas on best practices that can be used and also shows what the proposed system aims to improve.

The review covers both commercial and open-source LMS systems that are often used in schools, colleges, universities, and training centers. These systems are assessed based on factors like managing users, running courses, delivering content, handling assessments, keeping track of attendance, providing communication tools, generating reports, making sure they are accessible, how well they can grow, how customizable they are, and how user-friendly they are. Their suitability for smaller and medium-sized educational institutions is also considered.cIn addition to reviewing similar systems, this chapter includes a critical comparison of the pros and cons of existing solutions.

It also includes a background analysis and surveys conducted with people like administrators, teachers, and students to better understand the academic challenges faced by A TOO Z College. The insights gathered from these people help identify the functional and non-functional needs of the proposed Smart Learning Management System and ensure the developed solution meets the institution's real needs. The findings in this chapter provide the foundation for the project and support the choice of technologies, design decisions, and methods used during the system's development.

Additionally, the research gaps identified show the need for a tailored, lightweight, affordable, and responsive Learning Management System that meets the academic and administrative needs of A TOO Z College while providing a framework for future improvements.

Similar System 1 – Moodle

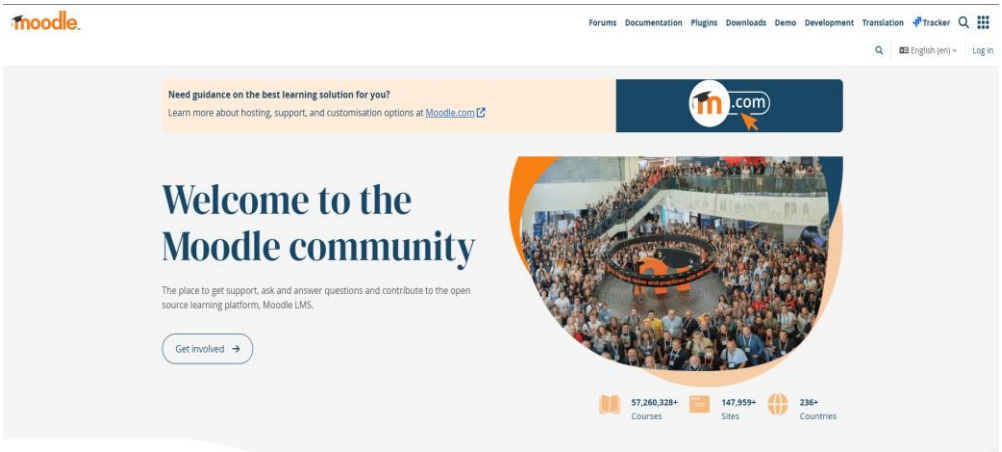


Figure 3: Moodle Dashboard Interface

Moodle is a popular open-source learning management system used worldwide. It was first created by Martin Dougiamas in 2002 with the goal of giving educators, administrators, and students a secure and flexible platform for online learning. Because it's open source, many schools, universities, and training centers use it.

Moodle offers a wide range of tools for managing courses, delivering content, communicating, assessing students, grading work, and keeping track of progress. Teachers can create courses and add materials like notes, videos, presentations, assignments, quizzes, discussion forums, and announcements from a single place. Students can access these materials, submit homework, take quizzes, talk to teachers, and check their grades through a web browser or the Moodle app on their phone.

One of Moodle's biggest strengths is how flexible it is. Since it's open source, developers can change how it works, add new features, and connect it to other tools as needed. It also has features like role-based access, multilingual support, learning analytics, grading tools, certificate creation, and detailed reports, making it useful for different types of education.

Moodle also has a strong community of users and developers who help improve the system. They create plugins, security updates, and learning materials, so institutions don't have to build everything from scratch. Moodle also follows modern e-learning standards like SCORM and LTI, which help it work with other educational tools and digital content.

However, Moodle has some disadvantages too. It needs technical knowledge for setup, configuration, and maintenance. Schools that don't have IT support might find it hard to manage. The interface can seem complicated for people who are new to it. Smaller schools with basic needs might find everything too complex because there are so many options and settings.

If Moodle is used on low-end servers or with many plugins, it can slow down. Regular updates, database management, and server maintenance are important for running it smoothly and securely. Customizing Moodle beyond its basic features usually needs experienced PHP developers who understand its structure.

Compared to Moodle, the proposed Smart Learning Management System (LMS) is designed for A TOO Z College. Instead of offering a wide range of advanced features for large universities, it focuses on the basic academic functions that the college needs. These include managing users, departments, subjects, courses, student enrollment, study materials, assignments, quizzes, attendance, announcements, messaging, and tracking academic performance through a simple and easy-to-use interface.

The Smart LMS also emphasizes ease of use, a lightweight design, responsive web layout, and simple administration. By keeping things straightforward while still having the tools needed for managing education, the system improves usability for teachers, students, and staff. It is designed to be practical and cost-effective, better matching the specific needs of A TOO Z College.

Advantages of Moodle

- It's open source and free to use.
- Can be customized with plugins and extensions.
- Supports course management, assignments, quizzes, grading, and discussion forums.
- Has strong community support and ongoing improvements.
- Offers multilingual support and mobile access.
- Provides detailed reports and learning analytics.
- Can be used by institutions of all sizes.

Limitations of Moodle

- Requires technical skills for setup and maintenance.
- The interface may be too complex for beginners.
- Performance depends on server quality and setup.
- Advanced customization needs experienced developers.
- Too many plugins can make maintenance more difficult.

Relevance to the Proposed Smart LMS Looking at Moodle helps us understand what features are important in a good learning management system. Ideas like user roles, course management, digital assessments, attendance tracking, and reporting have influenced the design of the Smart LMS. But unlike Moodle, the Smart LMS focuses on simplicity, ease of customization, maintenance, and responsiveness. This makes it better suited to the specific needs of A TOO Z College.

Similar System 2 – Google Classroom

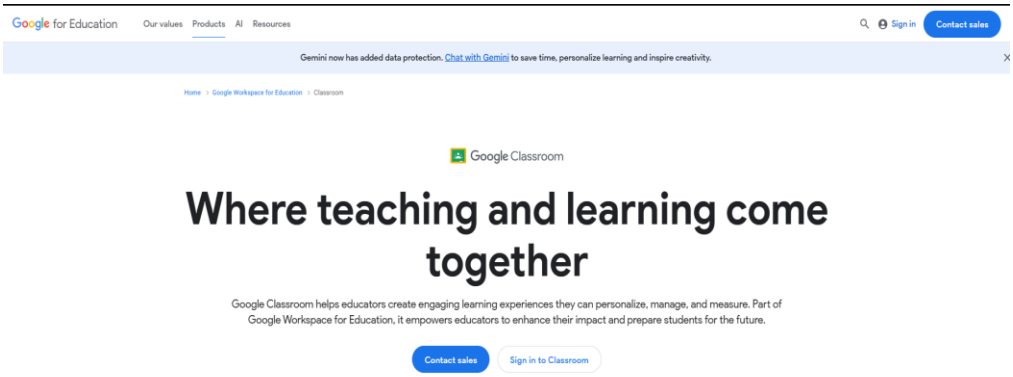


Figure 4: Google Classroom Dashboard

Google Classroom is a cloud-based Learning Management System made by Google to make teaching and learning easier. It works with other Google services like Google Drive, Google Docs, Google Meet, and Google Forms. This lets teachers give out learning materials, create assignments, talk to students, and handle coursework through a web browser or a mobile app.

Many schools and colleges use Google Classroom because it's easy to use and has a simple design. Teachers can make virtual classrooms quickly, share important messages, give homework, and check in on student work. Students can access learning resources, turn in assignments, get feedback, and chat with teachers using their Google accounts.

Even though Google Classroom is good for online learning and working together in class, it doesn’t offer all the tools needed for full academic management. Features like advanced attendance tracking, detailed reports on student performance, managing different departments, and handling institution-specific tasks are either missing or need extra apps from other companies.

Table 1table

Pros	Cons
Simple and easy-to-use interface	Limited academic administration features
Free for educational institutions	Requires Google accounts for all users
Integrates with Google Workspace applications	No built-in department or course management
Supports assignment submission and grading	Limited attendance tracking capabilities
Accessible via desktop and mobile devices	Limited customization for institutional requirements
Easy file sharing through Google Drive	Basic reporting and analytics compared to full LMS platforms

Research Gap

Google Classroom works well for sharing content and managing assignments, but it's not enough for A TOO Z College's full academic needs. The proposed Smart Learning Management System fills these gaps by bringing together all these features—user management, department management, subject management, course management, enrollment, attendance tracking, quizzes, performance monitoring, announcements, messaging, and detailed reporting into one web-based platform made specifically for the institution.

Similar System 3 – Canvas LMS

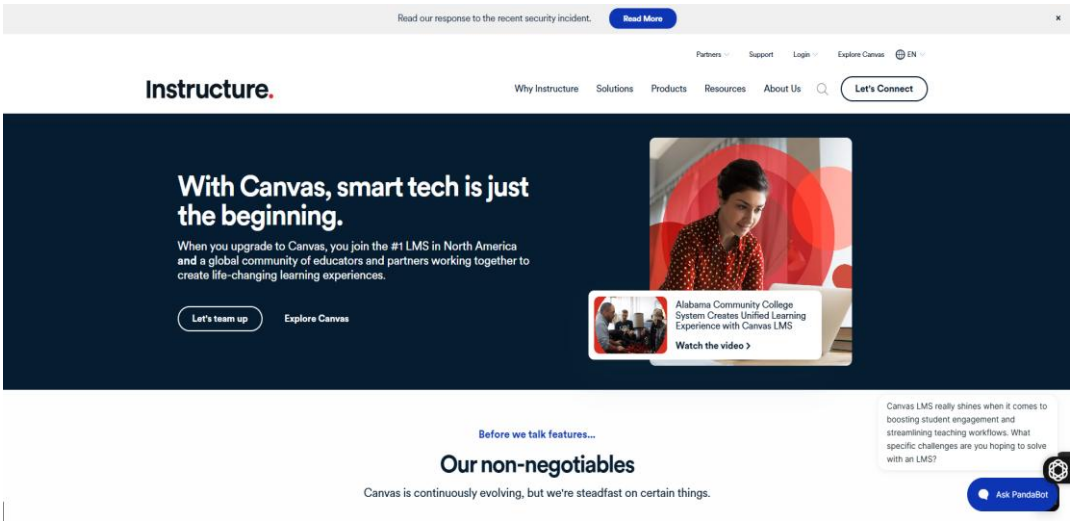


Figure 5: Canvas LMS Dashboard

Canvas LMS is a cloud-based Learning Management System created by Instructure. It is commonly used by universities, colleges, and educational institutions to support online, blended, and in-person learning. Canvas has a modern and easy-to-use interface that lets teachers manage courses, share learning materials, give assessments, and communicate with students in a smooth way.

The platform includes features like managing courses, submitting assignments, taking online quizzes, grading work, using discussion forums, setting up a calendar, and keeping track of student progress. It also allows integration with other educational tools through Learning Tools Interoperability (LTI), which helps institutions set up flexible and scalable learning environments.

While Canvas LMS has many useful academic features and is easy to use, it is mainly made for large educational institutions. The system usually needs a license or subscription fee for institutional use, which can be expensive for smaller

colleges. Also, some advanced settings and administrative tools need technical knowledge and support from the institution.

Table 2: Pros and Cons of Canvas LMS

Pros	Cons
Modern and user-friendly interface	Requires institutional licensing or subscription
Supports online, blended, and face-to-face learning	Expensive for small educational institutions
Strong assignment and quiz management	Some advanced features require training
Mobile application available	Limited customization compared to open-source LMS platforms
Supports third-party integrations (LTI)	Depends on cloud services and internet connectivity

Research Gap

Canvas LMS is a strong platform that provides complete learning management tools for higher education. However, its cost and design for large institutions make it not ideal for A TOO Z College. The proposed Smart Learning Management System is built as a cost-friendly, simple, and fully tailored solution that meets the specific academic and administrative needs of the institution. It includes key features like user management, attendance tracking, assignment submission, quiz

management, announcements, messaging, and academic reporting, all through a simple and responsive web interface.

Similar System 4 – Blackboard Learn

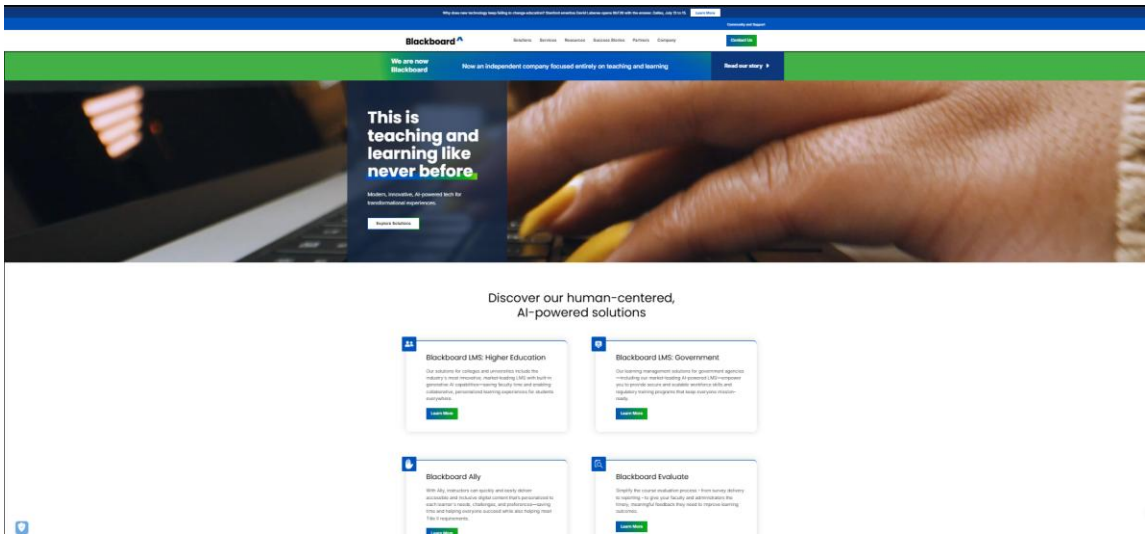


Figure 6 Blackboard Learn Dashboard

Learn Blackboard Learn is a Learning Management System made by Blackboard Inc. It is commonly used by universities, colleges, and big educational institutions to support online and blended learning. The platform gives a single place where teachers can handle their courses, share learning materials, do assessments, talk to students, and check how well students are doing.

Blackboard Learn includes features like course management, submitting assignments, taking online exams, grading work, having discussion forums, making announcements, keeping track of attendance, and using learning analytics. The system also works with many other educational tools, which makes it good for institutions that need a full academic management system.

Even though Blackboard Learn has a lot of features, it is usually made for large educational organizations that have enough money and technical support.
The system needs to be paid for, regularly updated, and users need training, which can make the whole setup more expensive. Plus, having so many features can make the system harder for users who only need basic functions.

Pros	**Cons**
Comprehensive academic management features	High licensing and implementation cost
Strong security and access control	Complex interface for new users
Supports online assessments and grading	Requires user training
Provides detailed reporting and learning analytics	Higher maintenance requirements
Suitable for large educational institutions	More features than required for small colleges
Integrates with various educational technologies	Requires reliable technical support

Table 3: Pros and Cons of Blackboard Learn

Research Gap

Blackboard Learn is a strong and dependable Learning Management System for big educational institutions. However, its high cost, complex setup, and advanced features make it unsuitable for A TOO Z College. The proposed Smart Learning Management System is designed to give the basic academic management features that the college needs through a simpler, more affordable, and easier-to-use web-

based platform. It focuses on user management, course handling, attendance tracking, submitting assignments, quizzes, announcements, messaging, and academic reports while keeping the system simple and reducing the cost of maintenance.

Similar System 5 – Schoology

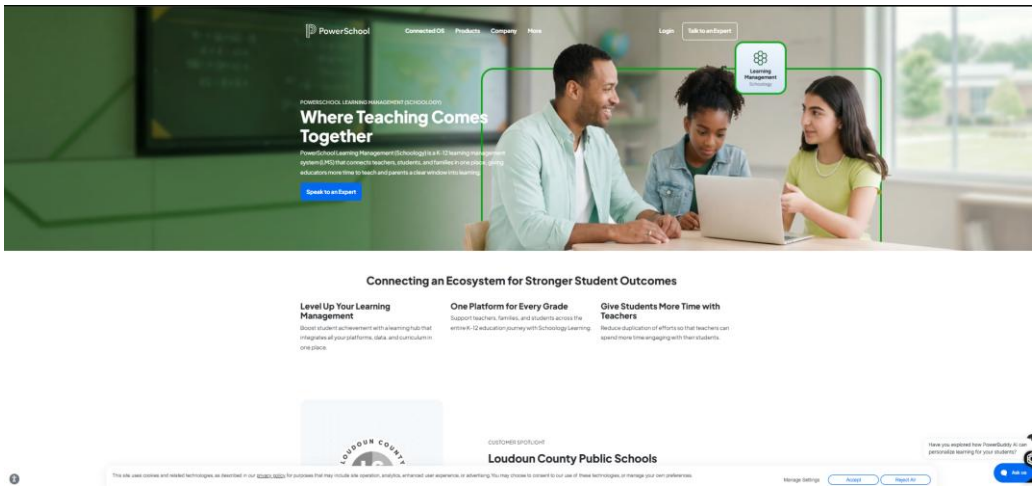


Figure 7: *Schoology Dashboard*

Schoology is a cloud-based Learning Management System that helps with digital teaching, learning, and classroom teamwork. It is popular among schools, colleges, and other educational places because it has a user-friendly design and strong communication tools. The platform allows teachers to build courses, share learning materials, give assignments, run quizzes, track grades, and talk to students all in one online space.

Schoology mixes learning management with social learning tools, letting students and teachers work together through discussion areas, news updates, messages, and shared resources. The platform is also available on mobile devices, so users can access course content and take part in academic activities anytime and from anywhere.

Even though Schoology has an easy-to-use interface and good collaboration features, some advanced academic management tools need a premium subscription. Also, the system doesn't offer a lot of customization options, which can be a problem for schools that have specific administrative needs or unique

academic processes. These institutions might need to use other tools or services to get full reporting and advanced management features.

Pros	Cons
User-friendly and intuitive interface	Some advanced features require a paid subscription
Strong communication and collaboration tools	Limited customization for institutional needs
Supports assignments, quizzes, and grading	Advanced reporting features are limited
Mobile application available	Some administrative functions require third-party integration
Easy course and content management	Less suitable for complex institutional management
Cloud-based with easy accessibility	Depends on stable internet connectivity

Table 3: Pros and Cons of Schoology

Research Gap

Schoology is a good platform for online teaching, teamwork, and classroom management. But it doesn't fully meet the specific academic and administrative needs of A TOO Z College. The proposed Smart Learning Management System is made to offer all the necessary features in one web-based platform, including user management, department management, subject management, course management, student enrollment, attendance tracking, assignment submission, quiz handling, announcements, messaging, and academic performance reports. This custom solution is simpler, more cost-effective, and tailored to the institution's needs, while still being easy to use and accessible on different devices.

Additional Similar Systems

In addition to the five main Learning Management Systems covered earlier, there are other LMS platforms that are commonly used in schools and businesses. These systems offer various combinations of tools for managing learning, communication, testing, teamwork, and administration. Even though they aren't

discussed in depth, they still give us a good idea of what's happening in digital learning and school management right now.

Microsoft Teams for Education

Microsoft Teams for Education is a cloud-based platform that helps with teamwork and learning. It connects with other Microsoft apps like Word, Excel, PowerPoint, OneDrive, and OneNote, letting teachers and students work together in one place. The platform includes virtual classrooms, submitting homework, video calls, file sharing, announcements, and chats.

Microsoft Teams is great for places that need real-time collaboration and online classes. However, many school-related tasks such as tracking attendance, managing departments, and handling institutional issues often need to be linked with other Microsoft tools or third-party apps.

Edmodo

Edmodo was a cloud-based platform that helped teachers, students, and parents communicate. It had features like managing assignments, taking quizzes, posting announcements, having discussions, and sharing resources. The interface was similar to social media sites, making it easy to use.

Edmodo became popular because of its user-friendly design and supportive learning environment. But it was shut down in 2022, which means it's no longer available for schools. Even though it's no longer active, it showed how social learning can make students more involved in online learning.

Chamilo LMS

Chamilo is an open-source LMS designed to offer affordable and flexible e-learning options for schools and training groups. It includes tools for managing courses, online assessments, assignments, learning paths, user management, certificates, and tracking progress.

Chamilo is known for its easy-to-use interface and lower hardware needs compared to other big LMS platforms. However, it has a smaller community and fewer plugins than Moodle, which means it doesn't offer as many customization options or extra features.

Open edX

Open edX is an open-source LMS originally made by Harvard University and MIT. It's designed for large online learning environments and MOOCs. It

provides tools like delivering multimedia content, interactive tests, discussion forums, tracking learner progress, and issuing certificates.

While Open edX can support many learners at once, it needs a lot of technical know-how to set up, configure, and keep running. Because of this, it works best for big universities and large educational organizations, not smaller ones.

Talent LMS

Talent LMS Is a cloud-based LMS mainly for workplace training, employee development, and professional education. It includes features like creating courses, tests, getting certifications, tracking learner progress, reporting, and mobile access. The platform is simple to use and maintain because everything is hosted in the cloud. However, it's more focused on training within organizations rather than handling all the day-to-day activities of a school, making it less suitable for universities.

System	Deployment	Primary Focus	Major Strength	Main Limitation
Microsoft Teams for Education	Cloud-Based	Online Collaboration	Excellent communication and video conferencing	Limited academic administration features
Edmodo	Cloud-Based	Classroom Communication	Simple and collaborative learning environment	Discontinued and no longer supported
Chamilo LMS	Open Source	Academic Learning	Lightweight and easy to use	Smaller community and fewer plugins
Open edX	Open Source	Large-Scale Online Learning	Highly scalable and feature-rich	Complex installation and maintenance
TalentLMS	Cloud-Based	Corporate Training	Easy deployment and mobile accessibility	Limited support for academic administration

Table 4: Comparison of Additional Similar Systems

Discussion

Looking at these other LMS platforms shows that each one is made to meet certain educational or business needs.
Some focus on teamwork and virtual learning, while others are designed for large-scale online education or company training. However, none of them fully meets all the needs of A TOO Z College without extra customization or linking with other systems. The proposed Smart Learning Management System is different because it's a custom-made, simple, and institution-specific solution. It brings together all the tools needed for managing academic activities, including course management, attendance, submitting assignments, quizzes, announcements, messaging, and tracking performance, all in one web-based app. This focused approach makes it easy to use, affordable to maintain, and well-suited for the day-to-day needs of the institution.

Background analysis and survey among stakeholders

The successful creation of a Learning Management System relies on a deep understanding of what its users need and expect. Because of this, a background study and survey were done to learn about the current academic practices, challenges, and system needs at A TOO Z College. The information gathered during this stage was used to build the functional and non-functional needs for the new Smart Learning Management System (LMS).

The main groups involved in this project are the college administrators, teachers, and students. These groups play a direct role in the everyday academic life of the college and are the main users of the system. Conversations and surveys were conducted with representatives from each group to better understand their duties, problems they face, and what they want from a digital learning platform.

2.2.1. Background Analysis

The analysis showed that the college currently uses many manual methods and various digital tools for academic tasks. Learning resources are shared through different platforms, attendance is recorded manually, assignments are collected in different ways, and academic data is stored separately. This scattered approach causes repeated work, communication delays, inconsistent records, and more work for administrators.

Lacking a centralized Learning Management System also limits the college's ability to track student performance, make academic reports quickly, and keep academic records safe. These problems become more obvious during online or blended learning, when constant digital access to educational resources is needed.

2.2.2. Survey among Stakeholders

To understand the system needs better, discussions and informal surveys were done with the main stakeholder groups.

Administrators

Administrators wanted a system that can manage users, departments, subjects, courses, enrollments, announcements, and reports. They also wanted to reduce manual work and improve the accuracy and access to institutional data.

Teachers

Teachers wanted a platform where they can upload study materials, create assignments and quizzes, take attendance, communicate with students, and track academic performance. They also wanted a simple interface to save time on routine tasks.

Students

Students wanted a single place where they can easily get learning materials, submit assignments, take quizzes, receive updates, check attendance, and see their academic results. They also wanted the system to be accessible on both mobile devices and computers.

Stakeholder	Current Challenges	Expected System Features
Administrators	Manual record management, separate systems, time-consuming reporting	User management, department management, reporting, announcements, academic records
Teachers	Difficulty managing learning materials, attendance, assignments, and communication	Course management, study material upload, attendance, quizzes, assignments, messaging
Students	Difficulty accessing resources, submitting assignments, and receiving updates	Course access, assignment submission, quizzes, attendance viewing, announcements, academic records

Table 5 Summary of Stakeholder Requirements

Key Findings

The background analysis and stakeholder surveys found several shared needs across all user groups. These include:

- A central platform for managing academic activities.
- Secure login systems that differ by user role.
- Digital management of courses, assignments, quizzes, and attendance.
- Efficient communication using announcements and messages.
- Accurate academic reports and tracking of student performance.
- A user-friendly interface that works on both desktops and mobile devices.
- A secure and reliable database for storing institutional data.

The findings from the requirement -gathering process helped shape the design and development of the proposed Smart Learning Management System. By meeting the needs of administrators, teachers, and students, the system aims to improve academic efficiency, reduce administrative work, and enhance the overall teaching and learning experience at A TOO Z College.

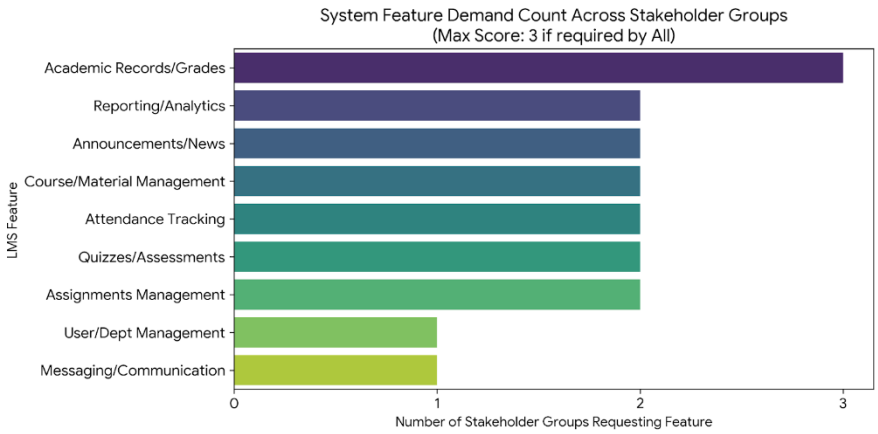


Figure 7: *Bar Chart of System Feature Demand Count Across Stakeholder Groups at A TOO Z College.*

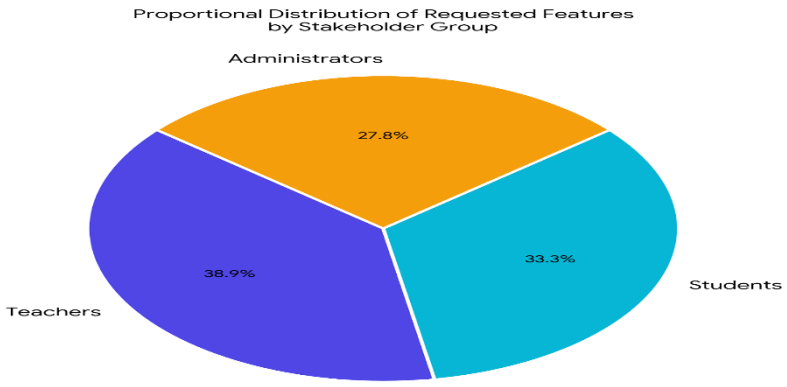


Figure 8: *Pie Chart Proportional Distribution of Requested LMS Features by Stakeholder Group.*

Critical evaluation of existing solutions

The review of current Learning Management Systems shows that modern LMS platforms have greatly improved how education is delivered and managed through digital tools. Systems like Moodle, Google Classroom, Canvas LMS, Blackboard Learn, Schoology, Microsoft Teams for Education, Chamilo, Open edX, and Talent LMS offer various features that support online learning, content management, assessments, communication, and student engagement. Each platform has been created to meet different educational needs, leading to differences in how they work, how complicated they are, how they are set up, and how much they cost.

Moodle is one of the most flexible and widely used open-source LMS. It can be customized a lot and has a big collection of plugins, which makes it a good fit for universities and institutions that have strong technical support. However, its complexity and the need for ongoing maintenance can be a problem for smaller schools that need a simpler solution.

Google Classroom is a simple and easy-to-use tool for managing classroom tasks and assignments. It works well with other Google apps, which makes it convenient for teachers and students. But it doesn't have many advanced features like full attendance tracking, managing different departments, or creating detailed institutional reports.

Canvas LMS has a modern look, good tools for assessments, and supports online learning well. While it provides a great learning experience, the cost of licensing and

its enterprise structure might not be ideal for schools with limited budgets and technical resources.

Blackboard Learn is a comprehensive LMS with strong security, analytics, and reporting. It works well for big universities but can be too complex and costly to set up and maintain, making it less practical for smaller institutions.

Schoology focuses on teamwork and classroom communication with an easy-to-use interface. It supports teaching and learning effectively, but it offers limited customization and requires paid subscriptions to unlock some advanced settings. Other systems like Microsoft Teams for Education, Chamilo, Open edX, Edmodo, and Talent LMS also offer useful features for specific educational or training settings.
But each system usually focuses on one area like collaboration, large online courses, corporate training, or cloud-based learning. They don't offer a fully customized solution for all the needs of A TOO Z College.

Based on this analysis, it’s clear that no existing LMS fully meets all the operational, academic, and administrative needs of A TOO Z College without major customization, extra software, or higher costs.

Most paid systems require licensing fees, while many free systems need lots of technical skills for installation, setup, and maintenance. The proposed Smart Learning Management System was created to address these issues. It is a custom-built, lightweight, and responsive web-based solution designed specifically for A TOO Z College. Unlike other systems, it includes everything needed for managing users, departments, subjects, courses, student enrollment, study materials, assignments, quizzes, attendance, announcements, messages, and academic performance all in one place. The system is designed to be simple, user-friendly, affordable, and easy to maintain. It also works on desktops, tablets, and smartphones.

The literature review shows that existing LMSs have provided helpful ideas and best practices that helped in designing the Smart LMS. However, the limitations found in these systems highlight the need for a tailored solution that can meet the specific educational and administrative challenges faced by A TOO Z College.

Evaluation Criteria	Moodle	Google Classroom	Canvas LMS	Blackboard Learn	Schoology	Proposed Smart LMS
Cost	Free (Open Source)	Free	Paid	Paid	Freemium	Low Development Cost
Ease of Use	Medium	High	High	Medium	High	High
Customization	High	Low	Medium	Medium	Low	High (Institution-Specific)
Attendance Management	Yes	Limited	Yes	Yes	Yes	Yes
Assignment Management	Yes	Yes	Yes	Yes	Yes	Yes
Quiz Management	Yes	Basic	Yes	Yes	Yes	Yes
Academic Reporting	Advanced	Basic	Advanced	Advanced	Moderate	Customized Reports
Department & Course Management	Yes	Limited	Yes	Yes	Limited	Fully Supported
Responsive Design	Yes	Yes	Yes	Yes	Yes	Fully Responsive
Suitability for A TOO Z College	Moderate	Moderate	Moderate	Low	Moderate	Excellent

Table 6 :: Overall Comparison of Existing Learning Management Systems

Conclusion of Critical Evaluation

The comparison shows that though existing LMSs have strong educational features, they are usually built for broader environments and may bring extra complexity, higher costs, or limited customization for A TOO Z College. Therefore, the proposed Smart LMS is a better fit because it brings together the essential features of academic management with simplicity, affordability, and customization that suits the specific needs of the institution.

2. 9 Summary

This chapter looked at the current research on Learning Management Systems (LMS) and looked at several popular platforms, like Moodle, Google Classroom, Canvas LMS, Blackboard Learn, Schoology, and others. Each system was studied to understand what it offers, its strengths and weaknesses, and how well it fits the needs of educational institutions. The review showed that while these systems have many tools for online learning and managing academics, they also have some issues. These include high costs to set up, complicated admin work, limited ability to change the system, and features that don't fit perfectly with the specific needs of A TOO Z College.

Looking closely at the reviewed systems, it became clear there was a gap in the current solutions that makes the development of a new Smart Learning Management System necessary.

The analysis showed that existing systems are usually built for larger educational setups and often need a lot of custom work to suit smaller colleges. These findings support the need for a simpler, more affordable, user-friendly, and tailored system that can bring together academic and administrative functions into one place.

The chapter also included background research and surveys of administrators, teachers, and students at A TOO Z College. The information collected during this requirement analysis helped identify the main challenges each group faces and led to the development of both functional and non-functional requirements for the new system. This confirmed the need for a centralized online platform that improves academic management, enhances communication, simplifies administrative tasks, and supports modern digital learning methods.

In summary, the literature review provided both the theory and practical basis for the proposed Smart Learning Management System. The knowledge from existing systems,

along with the needs identified through stakeholder feedback, directly shaped the design of the system's architecture, features, database structure, and how it will be built.

The next chapter, Chapter 3 – System Analysis and Design, goes into detail about the proposed system. It explains the overall system structure, what it needs to do, what it should be like in terms of performance and reliability, use cases, activity diagrams, database design, entity relationship diagrams, relational structure, and user interface layout that form the foundation for building the Smart Learning Management System.

CHAPTER 03

3. SYSTEM ANALYSIS AND DESIGN

Introduction

System analysis and design is a very important part of the Software Development Life Cycle (SDLC). In this stage, the needs that come from different people involved in a project are studied carefully and turned into a clear plan for how the system will work. This plan is the key that helps in building the system. A good system design ensures that the system does what it is supposed to do, and also makes it easier to use, keep up to date, grow, stay safe, and perform well.

This chapter explains how the Smart Learning Management System (LMS) was studied, planned, and prepared for building. It talks about the steps taken during the development process, the features the system needs to have, and the different parts of software, database, and equipment needed. It also shows how the system will be built using common tools and diagrams used in software engineering.

The Smart Learning Management System was created based on what A TOO Z College needed for their academic and administrative tasks. Information was collected from college staff, teachers, and students to understand what they needed and what challenges they face. These findings helped create a single online platform that supports managing academic work, learning, communication, keeping track of attendance, handling tests, and checking student performance.

To make sure the system is strong and easy to work with, the project followed a clear development process. This process includes finding out what is needed, analyzing those needs, designing the system, building it, putting parts together, checking it for problems, and evaluating its performance. Each step helps to improve the system and ensures that it works well for what it was meant to do.

This chapter also shows the structure of the system, the parts of software and database it uses, the hardware needed, and the tools used during development. These parts help show how the system will work together to support the school's operations. They also set the technical base for creating a safe, fast, and flexible Learning Management System.

The details in this chapter act as a guide for the next steps in building the system. Through careful study and thoughtful planning, the Smart Learning Management System is meant to help make academic work more efficient, improve communication between stakeholders, make administrative tasks easier, and support the move towards digital learning within the institution.

Proposed Methodology

The proposed Smart Learning Management System (LMS) was developed using the Incremental Software Development Model, a commonly used approach in software engineering. This method builds a system through smaller, manageable steps rather than creating the whole application at once. The system was broken down into smaller parts, with each part being designed, built, tested, and connected before moving on to the next. This way, the team could keep checking and improving the system throughout its development, which helped reduce risks and made sure each part worked correctly before it was added to the whole.

The Incremental Model was chosen because it works well for web-based systems with many separate parts. The Smart Learning Management System includes several key parts: user login, managing users, departments, subjects, courses, enrollment, study materials, assignments, quizzes, attendance, messages, and academic reports. These parts can be developed independently and then put together to form the complete system.

The development started with gathering and understanding the needs of the users. This involved talking to administrators, teachers, and students at A TOO Z College to find out what the system needed to do. All the information from this stage helped create the system design and plan.

Next, the system's structure, database setup, diagrams showing how the system works, and the layout of the user interface were designed. These designs helped guide the development of each part and kept everything consistent throughout the process.

The system was developed in small parts, one by one. The first parts included user authentication and admin management, followed by features for teachers and students, course management, attendance tracking, assignment submission, quizzes, announcements, messages, and reporting. Each part was tested by itself before being added to the main system. This step-by-step approach made it easier to find and fix problems, and it allowed for improvements to be made during the development.

Testing was done after each part was developed to make sure it worked as expected. The team checked each part individually, then tested how the parts worked together, and also made sure the system was user-friendly and handled data properly. Any problems found were fixed before moving on to the next part, which helped create a reliable and easy-to-maintain system. The Smart Learning Management System was built using HTML, CSS, Bootstrap 5, and JavaScript for the front end, PHP for the back end, and MySQL for the database. It was run on the XAMPP development environment. The system was designed to work on various devices, including desktops, laptops, tablets, and smartphones, to ensure it is easy to use anywhere.

In summary, the Incremental Software Development Model helped create the Smart Learning Management System in a clear and flexible way. It allowed the

team to build the system step by step, test each part as it was developed, and combine everything together to form a reliable system that meets the needs of A TOO Z College.

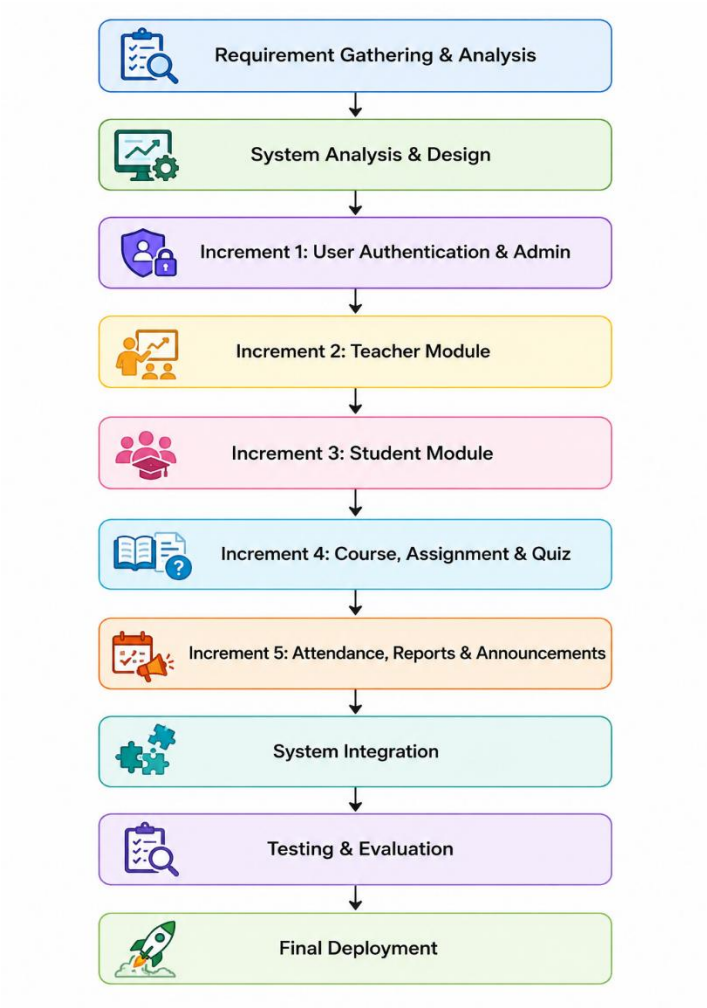


Figure 9: Proposed Incremental Development Methodology

3.2.1. The project is divided into distinct phases as follows:

The first step in the project was to collect and look closely at the needs for the new Smart Learning Management System (LMS). We talked with important people from A TOO Z College, like administrators,

teachers, and students. These talks helped us understand how the current academic processes work, what problems users are facing, and what features the new system should have.

Once we gathered all the information, we broke down the needs into two main groups: functional and non-functional. Functional needs included things like managing users, organizing courses, sharing study materials, submitting assignments, managing quizzes, tracking attendance, sharing announcements, and generating academic reports. Non-functional needs covered areas such as keeping data safe, making the system easy to use, ensuring it works well, being dependable, and being accessible on different devices.

The results from this stage gave us a clear picture of what users wanted and set the base for designing and building the proposed Smart Learning Management System.

3.2.2. Incremental Design

The Smart Learning Management System (LMS) was built using the Incremental Design method. In this approach, the system was broken down into different parts called modules. Each module was planned, made, checked, and put together one after another. Instead of making the whole system all at once, the team worked on each part step by step. This made the development process easier to manage, helped find and fix problems early, and made sure each part worked well before it was added to the system.

The first part developed was the User Authentication and Administrator Module. This allowed users to log in securely and gave different

access levels based on their roles. Once the administrator features were done, the Teacher Module was created. This helped teachers upload study materials, create assignments, manage quizzes, record attendance, and track student performance.

Next, the Student Module was added. It let students access courses, download materials, submit assignments, take quizzes, see attendance records, and read announcements. After the main user modules were complete, the system was expanded with features like Course Management, Assignment Management, Quiz Management, Attendance Management, Report Generation, and Announcement Management.

Each part was tested separately before being added to the full system. This helped catch and fix any issues early, which improved the overall quality and reliability of the LMS. The incremental method also allowed the team to keep checking and improving the system based on what was needed. This resulted in a stable, easy to use, and easy to update Learning Management System.

The Incremental Design method offered flexibility during development and made sure the final Smart Learning Management System met the needs of A TOO Z College. It also ensured that the software was of high quality and easy to maintain over time.

3.2.3. Implementation and Integration

After finishing the system design, the Smart Learning Management System (LMS) was built using modern web development tools. The front part of the app was made with HTML, CSS, Bootstrap 5, and JavaScript to create a responsive and easy-to-use interface. The back part was

made with PHP, and MySQL was used to store and manage all academic and administrative data. The system was built and tested on a local computer using the XAMPP server environment.

The development followed an incremental approach. Each part of the system was built one at a time, starting with user login and administrator features. Then came the teacher and student sections, along with course management, study material handling, assignment handling, quiz handling, attendance tracking, announcements, messaging, and academic reports.

Once each part was successfully built, it was connected to the rest of the system to form a complete web-based application. MySQL was used to link all parts, so that data could be shared and managed smoothly between different sections. Role-based access was added to make sure that only the right people could use the right features—like administrators, teachers, and students accessing only what they need.

During the linking process, the connection between the user interface, the system logic, and the database was checked carefully to make sure data moved correctly and the system worked reliably. Each section was tested after being linked to ensure that it worked well without causing issues with other parts of the system.

The whole process created a fully working Smart Learning Management System that acts as a main platform for managing academic activities, improving communication, and supporting digital learning at A TOO Z College.

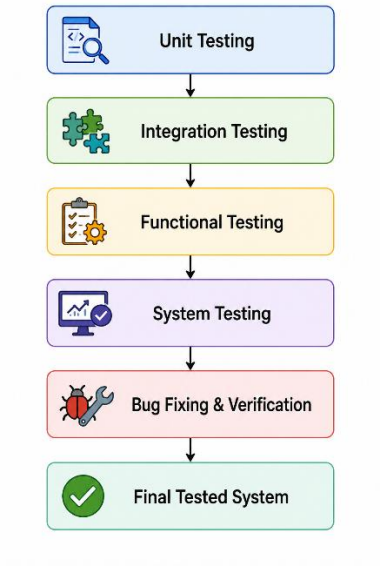


Figure 10: Testing Cycle of the Proposed Smart LMS

3.2.4. Testing Cycles

Testing was done all through the making of the Smart Learning Management System (LMS) to make sure every part worked well and met the set standards. Since the project used the Incremental Development Model, each part of the system was tested after it was completed before adding it to the main system. This way, problems were found and fixed earlier, which helped keep the final product free of major issues.

The testing started with Unit Testing, where each part of the system, like user login, course management, attendance tracking, submitting assignments, and managing quizzes, was tested separately to check if it worked as intended. Once each part was confirmed to work correctly, Integration Testing was done to ensure that all parts worked together smoothly and shared data properly through the MySQL database.

After the parts were successfully connected, Functional Testing was carried out to check if all the features of the system matched the requirements set during the planning stage. This testing covered important features like user login, access controls based on roles, managing courses, uploading study materials, submitting assignments, managing quizzes, recording attendance, sending announcements, messaging, and generating academic reports.

Finally, System Testing was done to check the whole Smart Learning Management System as a complete application. The system was tested for how easy it was to use, how dependable it was, how fast it responded, and how well it worked on different devices and web browsers. Any problems found during these tests were fixed before moving on to the next part of the development.

This ongoing testing helped improve the quality of the software, made sure the data was consistent, and confirmed that the Smart Learning Management System worked reliably and met the needs of A TOO Z College.

Requirements Analysis

Requirements analysis is an important step in building software. It helps find, study, and write down what the people using the system expect from it. This process makes sure that the final product meets user needs and supports the goals of the organization. The information gathered during this step gives a clear idea of what the system should do and how well it should work, which guides the design and building of the system.

For the Smart Learning Management System (LMS), we talked to key people at A TOO Z College, like administrators, teachers, and students, to understand their needs. We looked at how the college currently manages its academic activities, found out the problems they face, and decided what features would help improve teaching, learning, communication, and management tasks.

The requirements we collected fit into two groups: Functional Requirements and Non-Functional Requirements.
Functional requirements tell us what the system must do, like managing users, courses, attendance, assignments, quizzes, messages, and reports. Non-functional requirements explain how well the system should perform, including aspects like security, speed, reliability, ease of use, how easy it is to maintain, and how quickly it responds to user actions.

By carefully looking at all these requirements, we made sure the Smart Learning Management System would meet both the needs of A TOO Z College and the expectations of its users.

These requirements also helped us create the system's structure, database, user interface, and plan for building the system, which will be explained in the following sections of this report.

3.2.5. Functional Requirement

Functional requirements explain the services and tasks the Smart Learning Management System (LMS) must do to meet the needs of A TOO Z College. These requirements were gathered and studied during the

planning phase and are grouped based on the three main user types: Administrator, Teacher, and Student.

Administrator Functional Requirements

The Administrator is in charge of managing how the system works. The Administrator can:

- Make, change, or stop user accounts.
- Choose user roles like Administrator, Teacher, or Student.
- Handle departments.
- Manage subjects.
- Handle courses.
- Assign teachers to courses.
- Manage student enrollments.
- Share system-wide notices.
- Check attendance reports.
- Check student performance reports.
- Keep academic records.
- Update user details and system settings.

Teacher Functional Requirements

Teachers are responsible for managing learning activities and checking student work. The Teacher can:

- Log in securely using role-based authentication.
- See the courses they are assigned to.
- Upload study materials.
- Create and manage assignments.
- Create and manage quizzes.
- Record student attendance.
- View past attendance.
- Share course notices.
- Send and receive messages.
- Grade assignment submissions.
- Keep track of student performance.

Student Functional Requirements

Students use the system to access learning material and take part in classes. The Student can:

- Log in securely using role-based authentication.
- See the courses they are enrolled in.
- Download study materials.
- Submit assignments.
- Take online quizzes.
- Check attendance records.
- See their academic performance.
- Get course announcements.
- Ask questions or send messages.
- Update personal information.

User Role	Main Functional Requirements
Administrator	User management, department management, subject management, course management, teacher assignment, enrollment management, announcements, attendance reports, academic reports
Teacher	Course management, study material upload, assignment management, quiz management, attendance recording, announcements, messaging, student performance monitoring
Student	Course access, study material download, assignment submission, quiz participation, attendance viewing, academic performance viewing, announcements, messaging

Table 7 : Summary of Functional Requirements

The above functional requirements show the main services that the Smart Learning Management System provides. These requirements helped shape the design and build of each part of the system, making sure it supports the academic and administrative needs of A TOO Z College in a helpful and efficient way.

3.2.6. Non-Functional Requirements

Non-functional requirements explain the quality and working features of the Smart Learning Management System (LMS). These requirements help make sure the system is safe, dependable, works quickly, and is easy to use for everyone involved, like administrators, teachers, and students.

Security

The system should allow users to log in with different levels of access based on their roles, such as administrators, teachers, and students. Important user information and academic records must be kept safe and not accessed by anyone who isn't allowed. Only users who have been properly checked and given access can use system resources according to their roles.

Performance

The system should respond quickly to user actions like logging in, accessing courses, submitting assignments, recording attendance, and generating reports. It should handle many users without slowing down or causing problems during regular use.

Reliability

The Smart Learning Management System should work well and not have many problems. The data stored in the MySQL database needs to be correct and not change unexpectedly. The system should also back up the database regularly to avoid losing important information.

Usability

The system should have a clean, simple, and easy-to-use interface that everyone, including administrators, teachers, and students, can use without much difficulty. It should have a clear and consistent layout to help new users learn how to use it more easily.

Scalability

The system needs to be built in a way that allows for future growth, such as adding new academic features, handling more users, and connecting with new technology, without needing big changes to the existing system.

Maintainability

The application should be built using a modular design, so that parts of the system can be changed, fixed, or improved without affecting the rest of the system. Using good coding practices and organizing the database properly helps make the system easier to maintain over time.

Availability

The system should be available whenever the web server is running, so users can access academic materials and services anytime through an internet or local network connection.

Compatibility

The Smart Learning Management System should work on major web browsers like Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. It should also be designed to function well on different devices, such as desktops, laptops, tablets, and smartphones.

Requirement	Description
Security	Role-based authentication and secure access to academic data.
Performance	Fast response time and efficient processing of user requests.
Reliability	Stable system operation with accurate and consistent data management.
Usability	User-friendly interface with simple navigation.
Scalability	Supports future expansion and additional system modules.
Maintainability	Modular architecture for easier updates and maintenance.
Availability	Accessible whenever the server is operational.
Compatibility	Supports multiple browsers and responsive devices.

Table 8: Summary of Non-Functional Requirements

The above non-functional requirements help ensure that the Smart Learning Management System not only works as it should but also provides a safe, dependable, fast, and easy-to-maintain platform that supports the academic and administrative needs of A TOO Z College.

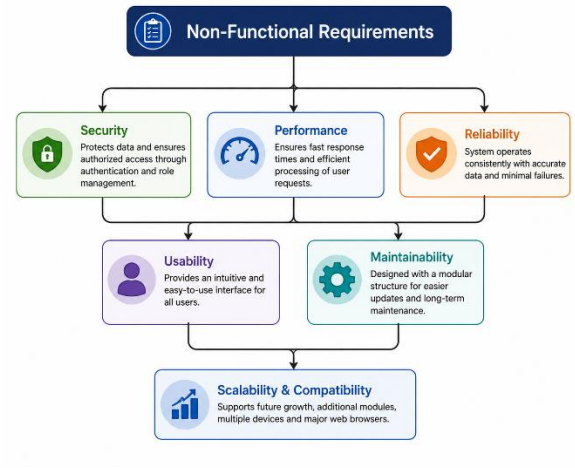


Figure 11: Non-Functional Requirements Overview

Components of the System

The Smart Learning Management System (LMS) is built using a modular structure that includes software, database, and hardware parts working together to create a secure, dependable, and efficient platform for managing academic activities. Each part has its own job in making the system work properly and helping users interact with the application smoothly.

The software part is in charge of the user interface, the logic that runs the system, and the processing that happens on the server. It lets administrators, teachers, and students do their tasks related to education and college management through a website that works well on different devices.

The database part acts as a main storage place for all the information used by the system. This includes user profiles, departments, subjects, courses, student enrollments, study resources, homework, tests, attendance records, news, messages, and results. It makes sure that data stays accurate, consistent, and safe while the system is running.

The hardware part provides the necessary computer resources to run the application. Devices like desktops, laptops, tablets, and phones let users access the system via a web browser. On the other side, the server setup runs the application, handles user requests, and manages database operations.

These parts are connected using up-to-date web technologies to make sure communication between the user interface, the system's logic, and the database works well. Together, they form a structure that can grow and be easily maintained, making it suitable for meeting the academic and administrative needs of A TOO Z College. The following sections explain each part used in making the Smart Learning Management System.

3.2.7. Software Component

The software part of the Smart Learning Management System (LMS) uses different technologies and tools to create, build, and run the web-based application. These software parts work together to make a secure, fast, and easy-to-use platform for people like administrators, teachers, and students.

The front-end of the system was made using HTML5, CSS3, Bootstrap 5, and JavaScript. HTML5 gives the structure to web pages, while CSS3 makes the app look better and more organized. Bootstrap 5 helps the system look good on all devices, including desktops, laptops, tablets, and phones. JavaScript adds interactive features like checking forms, updating content on the fly, and improving the user experience.

The back-end of the system was developed using PHP. PHP helps process user requests, manage the logic behind the app, handle user login and access, and

send and receive data from the database. It also supports important system tasks like managing users, courses, submissions, attendance, and reports.

All the data from the system is stored in a MySQL database. MySQL keeps track of everything related to the institution, such as user profiles, departments, subjects, courses, enrollments, study materials, quizzes, attendance, notices, and messages. It keeps the data safe, consistent, and makes it easy to find when needed.

The system is built and tested using XAMPP, which includes the Apache web server, PHP, and MySQL. XAMPP lets developers work on the app locally, test it thoroughly, and prepare it for use on a live server.

Software	Purpose
HTML5	Defines the structure of web pages.
CSS3	Styles the user interface and improves visual presentation.
Bootstrap 5	Develops a responsive and mobile-friendly interface.
JavaScript	Provides interactivity, validation, and dynamic functionality.
PHP	Implements server-side logic and business processes.
MySQL	Stores and manages all system data.
XAMPP	Provides the local development and testing environment.

Table 9Software Components Used in the Smart Learning Management System

Together, these software technologies create a reliable, scalable, and affordable system for the Smart Learning Management System. They also make it easier to update and maintain the system in the future. This mix of software components ensures the Smart Learning Management System is fast, secure, easy to manage, and can effectively support both teaching and administrative tasks at A TOO Z College.

The web-based dashboard of the system is accessible for admin via a web browser. Responsibility of the admin is managing the system features and update. The backend server handles the processing of data and communication between the mobile app, dashboard, and database.

3.2.8. Database Component

The database is a key part of the Smart Learning Management System (LMS). It handles all the data needed for academic and administrative tasks, including storing, organizing, finding, and managing this data. MySQL was chosen for the database because it offers good performance, security, data accuracy, and efficient management for web-based systems.

The database was built using the rules of Relational Database Management Systems (RDBMS) to reduce repeated data and keep everything consistent by using primary keys, foreign keys, and linking tables. During the design phase, an Entity Relationship Diagram (ERD) and a Relational Schema were made, which served as the basis for creating the database structure.

The Smart Learning Management System database has several connected tables that handle users, departments, teachers, students, subjects, courses, enrollments, study materials, assignments, quizzes, attendance, announcements, messages, and academic

records. These tables are linked using foreign key constraints, which help maintain accurate and consistent data across the system. Whenever users do things like logging in, adding study materials, submitting assignments, recording attendance, taking quizzes, or posting announcements, the PHP application handles the data and saves it into the MySQL database. The database also makes it easy to get information for creating reports, monitoring academic progress, and handling day-to-day administrative tasks.

The database performs several important functions:
- Keeping track of user account details.
- Managing departments, teachers, and students.
- Maintaining information about subjects and courses.
- Recording which students are enrolled in courses.
- Storing study materials uploaded by teachers.
- Managing assignments and the submissions of those assignments.
- Keeping quiz details. - Tracking attendance.
- Storing announcements and internal messages.
- Maintaining academic records and generating reports.

The database was built using MySQL in the XAMPP development environment. Primary keys are used to uniquely identify each entry, while foreign keys connect related tables, ensuring that data remains consistent and invalid records are prevented. This well-organized database design allows the Smart Learning Management System to work efficiently and also makes it easier to expand and update in the future.

3.2.8.1.Entities

No	Entity Name
1	Users
2	Admin

3	Teacher
4	Student
5	Department
6	Subject
7	Course
8	Enrollment
9	Study Material
10	Assignment
11	Assignment Submission
12	Quiz
13	Attendance
14	Announcement
15	Message

Table 10Main Entities in Smart LMS Database

3.2.8.2. Relational Schema

Entity	Primary Key	Attributes
Users	`user_id`	username, password, full_name, email, phone, role, status, created_at
Admin	`admin_id`	user_id (FK), designation, office_contact
Teacher	`teacher_id`	user_id (FK), department_id (FK), qualification, experience_years, joined_date

Student	`student_id`	user_id (FK), department_id (FK), registration_number, enrollment_year, status
Department	`department_id`	department_name, description, status, created_at
Subject	`subject_id`	department_id (FK), subject_code, subject_name, description, status
Course	`course_id`	subject_id (FK), teacher_id (FK), course_title, course_description, semester, academic_year, status, created_at
Enrollment	`enrollment_id`	student_id (FK), course_id (FK), enrollment_date, status
Study Material	`material_id`	course_id (FK), uploaded_by (FK), title, description, file_name, file_path, file_type, upload_date, status
Assignment	`assignment_id`	course_id (FK), teacher_id (FK), title, description, file_path, due_date, total_marks, status, created_at
Assignment Submission	`submission_id`	assignment_id (FK), student_id (FK), submission_file, submission_text, submission_date, marks, feedback, status
Quiz	`quiz_id`	course_id (FK), teacher_id (FK), quiz_title, description, time_limit, total_marks, start_time, end_time, status, created_at
Attendance	`attendance_id`	course_id (FK), student_id (FK), teacher_id (FK),

		attendance_date, status, remarks, created_at
Announcement	`announcement_id`	user_id (FK), course_id (FK), title, message, created_date, status
Message	`message_id`	sender_id (FK), receiver_id (FK), message_text, sent_date, read_status

Table 11 Relational Schema Entities and Attributes

3.2.8.3.Entity Relationships

Parent Entity	Child Entity	Relationship
Users	Admin	One-to-One (1:1)
Users	Teacher	One-to-One (1:1)
Users	Student	One-to-One (1:1)
Department	Teacher	One-to-Many (1:M)
Department	Student	One-to-Many (1:M)
Department	Subject	One-to-Many (1:M)
Subject	Course	One-to-Many (1:M)
Teacher	Course	One-to-Many (1:M)
Student	Enrollment	One-to-Many (1:M)
Course	Enrollment	One-to-Many (1:M)
Course	Study Material	One-to-Many (1:M)
Course	Assignment	One-to-Many (1:M)
Course	Quiz	One-to-Many (1:M)
Course	Attendance	One-to-Many (1:M)
Assignment	Assignment Submission	One-to-Many (1:M)
Users	Announcement	One-to-Many (1:M)
Users	Message	One-to-Many (1:M)

Table 12 Relational Schema Entities and Attributes

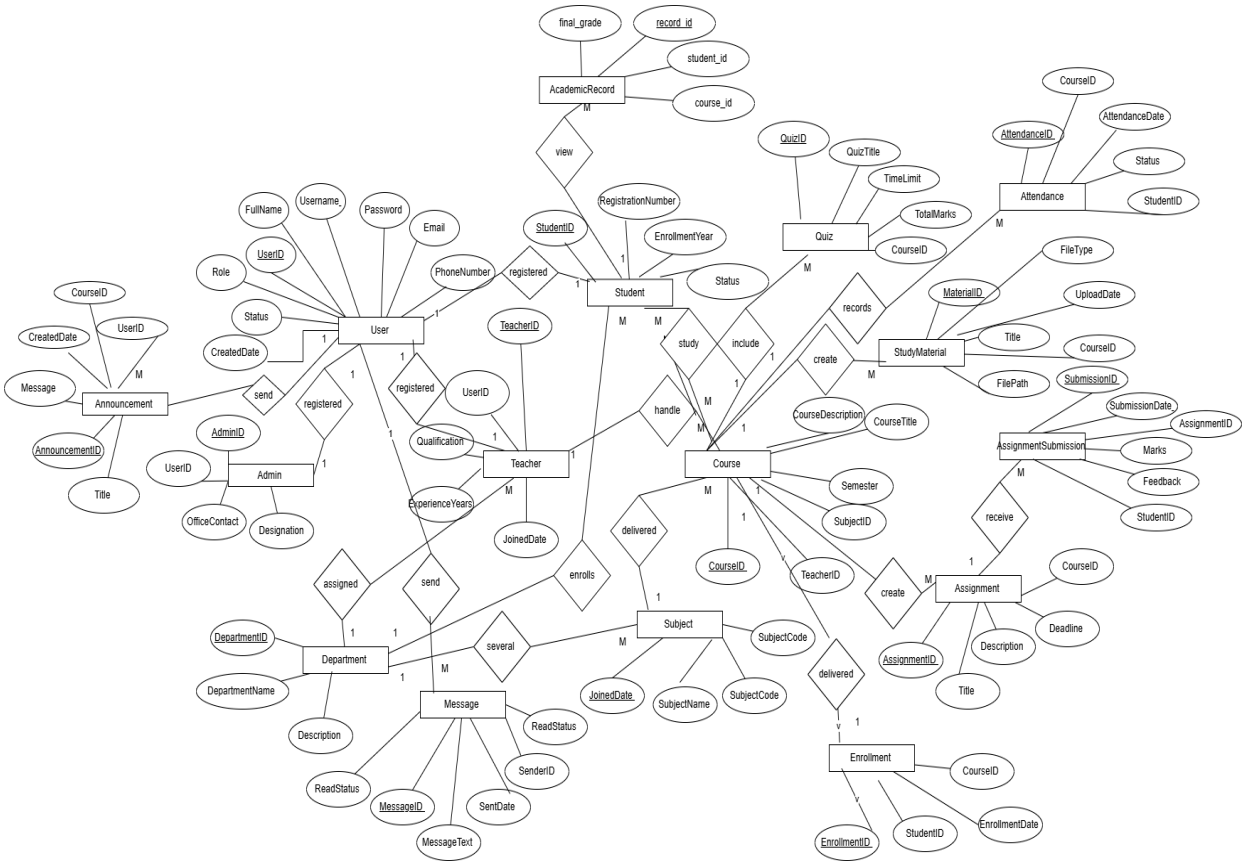


Figure 22: Entity Relationship Diagram (ERD)

3.2.8.4. DB diagram

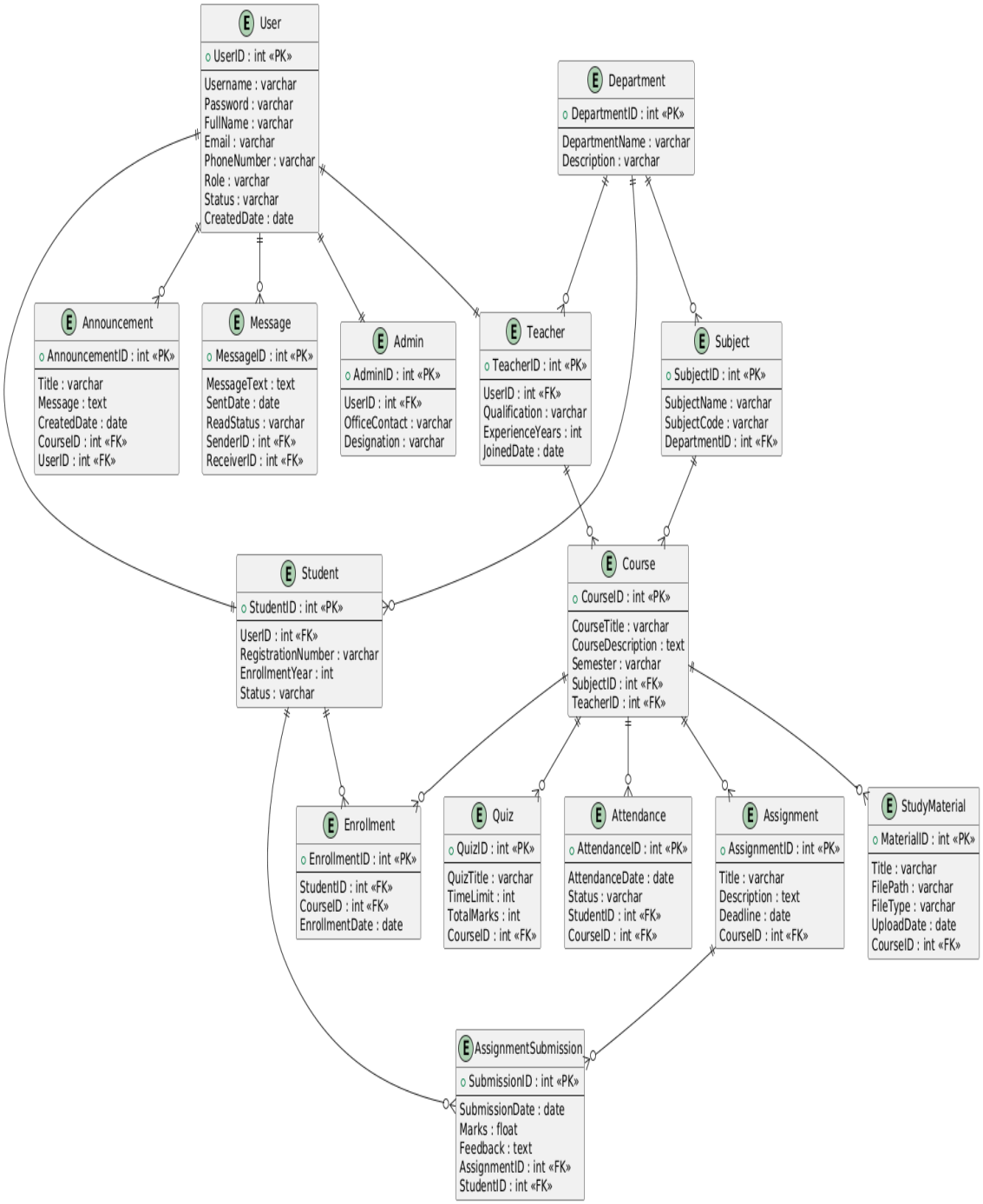


Figure 23: DB diagram

The Entity Relationship Diagram shows how the different parts of the database are connected. The Relational Schema shows the actual structure of the database in MySQL. Together, they help the Smart Learning Management System keep data accurate, retrieve information quickly, and support all the academic and administrative needs of the system.

Assumptions:

- The design and setup of the Smart Learning Management System (LMS) are based on these assumptions:

- Everyone who uses the system—like administrators, teachers, and students—has a valid account created by the system administrator.

- All users need a device such as a computer, laptop, tablet, or smartphone with a modern web browser and a working internet or local network connection.

- Administrators are responsible for keeping all information about departments, subjects, courses, teachers, and student details up to date and accurate.

- Teachers are expected to upload study materials, assignments, quizzes, and attendance records on time and in the correct way as per the academic schedule.

- Students must use their registered accounts to submit assignments, take quizzes, and access learning resources.

- The MySQL database must be running smoothly and able to store and retrieve data without major problems.

- The XAMPP server setup must be done correctly during the development and testing of the system.

- Only authorized users should follow the system's security rules and not share their login details with others who aren't allowed to use the system.

- Regular backups of the database must be done to avoid losing data and to keep all information secure and accurate.

- The institution must have the right hardware and software tools to properly install and run the Smart Learning Management System.

These assumptions help set the conditions under which the system is expected to work well. If any of these assumptions are not met, it might cause issues with how the system performs, stays secure, or runs reliably.

3.2.9. Hardware Component

The hardware part gives the physical setup needed for the Smart Learning Management System (LMS) to work well. It includes the devices that administrators, teachers, and students use to access the system, like computers, tablets, and phones. It also includes the servers that handle requests and keep the data safe. A strong hardware setup helps the system run smoothly, process data quickly, and always have access to learning materials.

The Smart Learning Management System is a web-based app that can be used through regular web browsers on desktops, laptops, tablets, and smartphones. Because it's responsive, users can do school and administrative tasks on any device without needing to install extra software.

During the development stage, the system was set up locally using XAMPP. This setup includes PHP for the app, Apache for the web server, and MySQL for the database, which helps connect the user interface with the data. In a real-world setup, the app can be placed on a dedicated server or a cloud hosting service to handle many users at the same time.

The system's hardware structure follows a client-server model. The devices used by users connect to the web server via the internet. The server then handles the user's requests, runs the necessary actions, gets data from the database, and sends it back to the user. This model helps manage data from one place, makes it more secure, and uses resources more efficiently. The next sections explain the devices used by users,

the server setup, and future plans for adding more hardware to the Smart Learning Management System.

Component	Description
Client Devices	Desktop computers, laptops, tablets, and smartphones used to access the LMS.
Web Server	Hosts the PHP application using Apache provided by XAMPP or a production web server.
Database Server	MySQL database server used to store academic and administrative information.
Network	Internet or Local Area Network (LAN) used for communication between clients and the server.

Table 13 Relational Schema Entities and Attributes

Technologies Adapted

The Smart Learning Management System (LMS) was built using up-to-date web technologies that ensure it works well, is easy to change, and simple to keep running. These technologies support making web apps that look good on any device, manage data efficiently, and process information securely. They were chosen because they are commonly used in web development, free to use, and fit the needs of A TOO Z College.

The front part of the system was made using HTML5, CSS3, Bootstrap 5, and JavaScript, which together create a web interface that looks good and works well on different screen sizes. The back part was made using PHP, while MySQL was used as the database to store and manage information.

HTML5 was used to structure the web pages. It helps set up the layout, forms, menus, tables, and other parts that the app needs.

CSS3 was used to make the system look good. It helps with colors, fonts, spacing, layouts, and animations, making the overall experience better for users.

Bootstrap 5 was used to create a responsive design that works on all devices. It provides ready-made parts like menus, buttons, forms, and layouts, which saves time and makes the design consistent across devices.

JavaScript makes the app interactive. It handles form checks, updates content on the fly, validates user input, and adds features that make the app easier to use.

PHP was picked for the back end because it helps connect the user interface with the MySQL database. It handles user login, manages sessions, performs tasks like adding, editing, and deleting data, and runs the logic that makes the app work.

MySQL was used as the database to keep all the academic and administrative data safe. It manages users, departments, teachers, students, subjects, courses, enrollments, assignments, quizzes, attendance, announcements, and messages, while making sure the data is accurate and consistent.

XAMPP was used as the local development tool during the build and testing. It brings together the Apache web server, PHP, and MySQL into one package, making it easier to develop and test the app before putting it live.

Using these technologies together creates a secure, flexible, and scalable system for the Smart Learning Management System. They work well together, allowing the app to communicate smoothly between the user interface, the code, and the database, and they also support future improvements and long-term care for the system.

Technology	Category	Purpose
HTML5	Front-End	Web page structure
CSS3	Front-End	User interface styling
Bootstrap 5	Front-End Framework	Responsive design
JavaScript	Client-Side Scripting	Dynamic functionality and validation
PHP	Server-Side Programming	Business logic and database communication
MySQL	Database	Data storage and management

Apache	Web Server	Hosts the web application
XAMPP	Development Environment	Local server for development and testing

Table 14: Technologies Used in the Smart Learning Management System

Explain Technology :
Frontend Technologies
HTML5, CSS3, JavaScript: For developing the responsive and user-friendly interface of the LMS.
Bootstrap 5: To ensure consistent responsive design across devices.
Socket. IO (Client Library): Enables real-time message exchange and notifications within the browser.

Backend Technologies
PHP: For core LMS functionalities such as authentication, assignment management, attendance, and academic records.
Node. js (JavaScript Runtime): Dedicated real-time chat microservice handling
WebSocket connections for instant communication.
Express. js Framework: Used with Node. js for managing chat API routes and
socket connections efficiently.
Socket. IO (Server Library): Provides reliable bi-directional real-time communication between server and clients (teachers and students).

Database
MySQL: Central database for storing user profiles, course data, chat messages, and message metadata (timestamps, read status, delivery status).

Additional Tools & Services
Firebase Cloud Messaging (Optional): For push notifications on web and mobile platforms.
JSON Web Tokens (JWT): For secure user authentication between PHP backend and Node. js chat service.
AWS (Amazon Web Services) or Local Hosting: Ensures scalable deployment and reliable uptime for both LMS and chat server.
Version Control: Git/GitHub for code management and collaborative development.

Figure 3.29: Technologies Used in the Smart Learning Management System

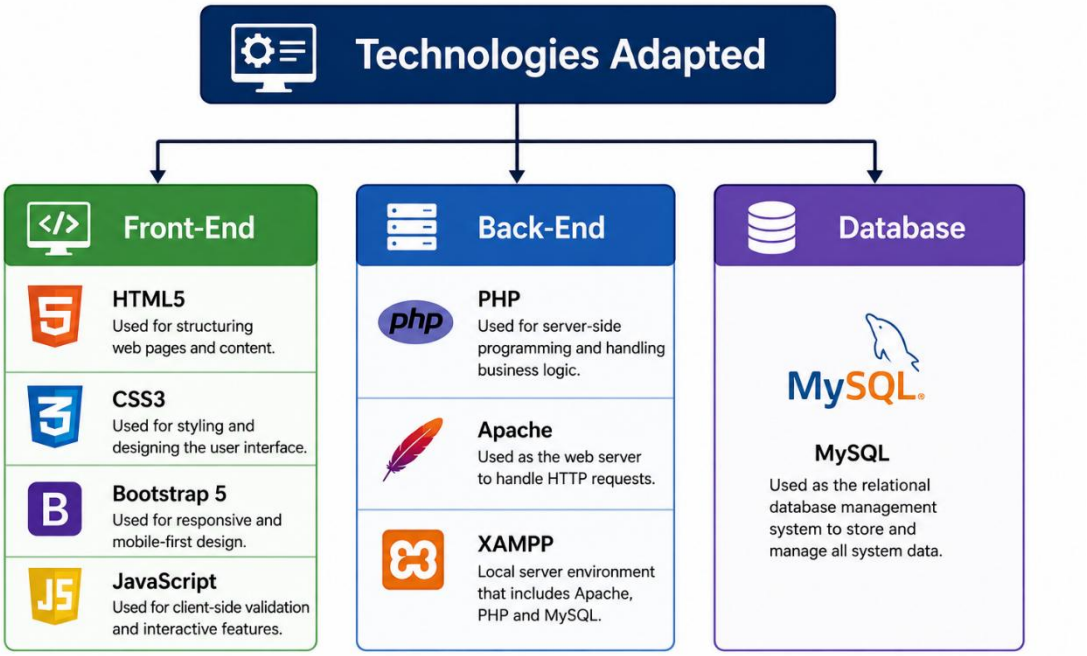


Figure 24: Technology Adopt

System Design

Design is the process of turning the system requirements into a clear plan that helps build the software application. It explains how the different parts of the Smart Learning Management System (LMS) work together to meet the needs identified during the analysis phase, both in terms of what the system should do and how it should perform.

The system design for the proposed Smart Learning Management System was made using standard software engineering methods to ensure it is modular, easy to maintain, scalable, and efficient in managing data. The design aims to deliver a secure, fast, and easy-to-use web application that supports the academic and administrative tasks of A TOO Z College.

The system uses a client-server structure. Users access the application through a web browser, and the server processes the requests using PHP. It stores all the data in a MySQL relational database. The system also provides role-based access, which means that administrators, teachers, and students can only access the features that are relevant to their roles.

During this phase, several software engineering models and design documents were created to fully represent the system. These include the Use Case Diagram, Activity Diagram, Entity Relationship Diagram (ERD), Relational Schema, Component-Based Architecture Diagram, and User Interface Design. Together, these models show how the system works, how users interact with it, how data is connected, how tasks are handled, and what the overall structure of the system is. The system design also focuses on making the website work well on different types of devices, including desktops, laptops, tablets, and smartphones, without affecting how it looks or works. This ensures that teaching, learning, and administrative tasks can be done efficiently no matter what device someone is using.

The following subsections describe the main design documents created for the proposed Smart Learning Management System. These diagrams give a clear picture of the system's structure, how it operates, how the database is organized, and how users and system parts interact before the system is built.

3.2.10. Use Case Diagram

The Use Case Diagram shows how users interact with the Smart Learning Management System (LMS). It gives an overview of the system by showing the main users, what they can do, and the features they can use. In this project, the main users are the Administrator, Teacher, and Student.

The Administrator has the most access and is in charge of managing the entire system. The Teacher uses the system to handle tasks like uploading study materials, creating assignments, giving quizzes, keeping track of attendance, and talking to students. The Student uses the system to get learning materials, submit assignments, take quizzes, see attendance records, check their grades, and read announcements.

The Use Case Diagram also includes common tasks like logging in and out, and controlling access based on user roles. These help ensure that each user can only use features that are allowed for their role. This makes the system safer and stops people from using parts they shouldn't.

Shows how the three main roles interact with the system's main features. It shows how the system handles academic management, teaching, learning, communication, assessment, attendance, and reporting. The Use Case Diagram helps define the system's limits and makes clear what each user is responsible for.

It also helps in designing system parts by showing which features are needed for each user role.

Actor	Description	Main Use Cases
Administrator	The administrator manages the overall LMS and has full control over system configuration and academic administration.	Manage users, manage departments, manage students, manage teachers, manage subjects, manage courses, manage enrollments, post announcements, view attendance reports, view performance reports
Teacher	The teacher manages course-related learning activities and student assessment tasks.	Upload study materials, create assignments, create quizzes, mark attendance, send announcements, view

		student performance, send and receive messages
Student	The student accesses learning resources and participates in academic activities through the system.	Access courses, download study materials, submit assignments, attempt quizzes, view attendance, view academic records, receive announcements, send academic queries

Table 15: Actor Explanation Table

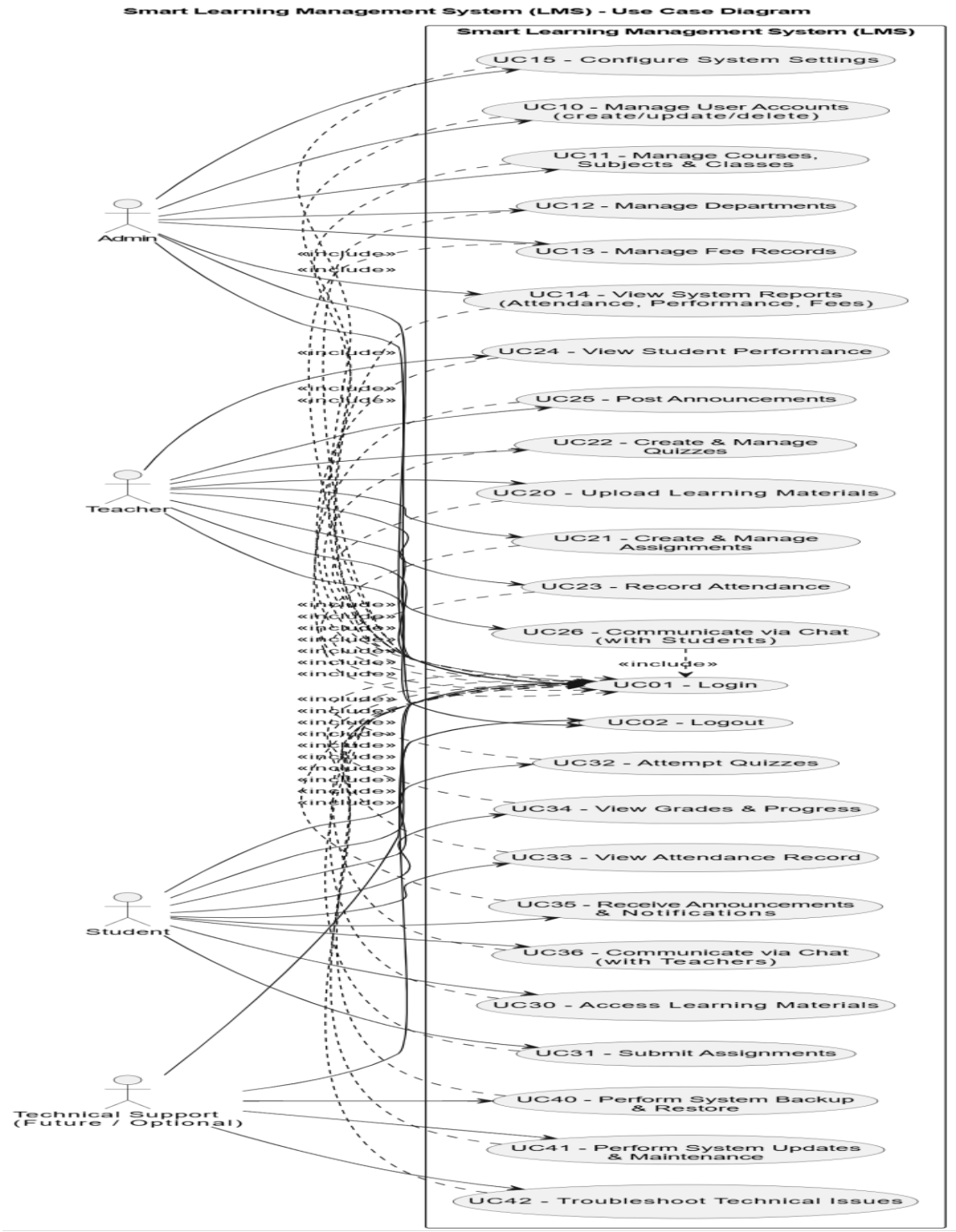


Figure 7 Use case for Overall System

3.2.11. Class Diagram

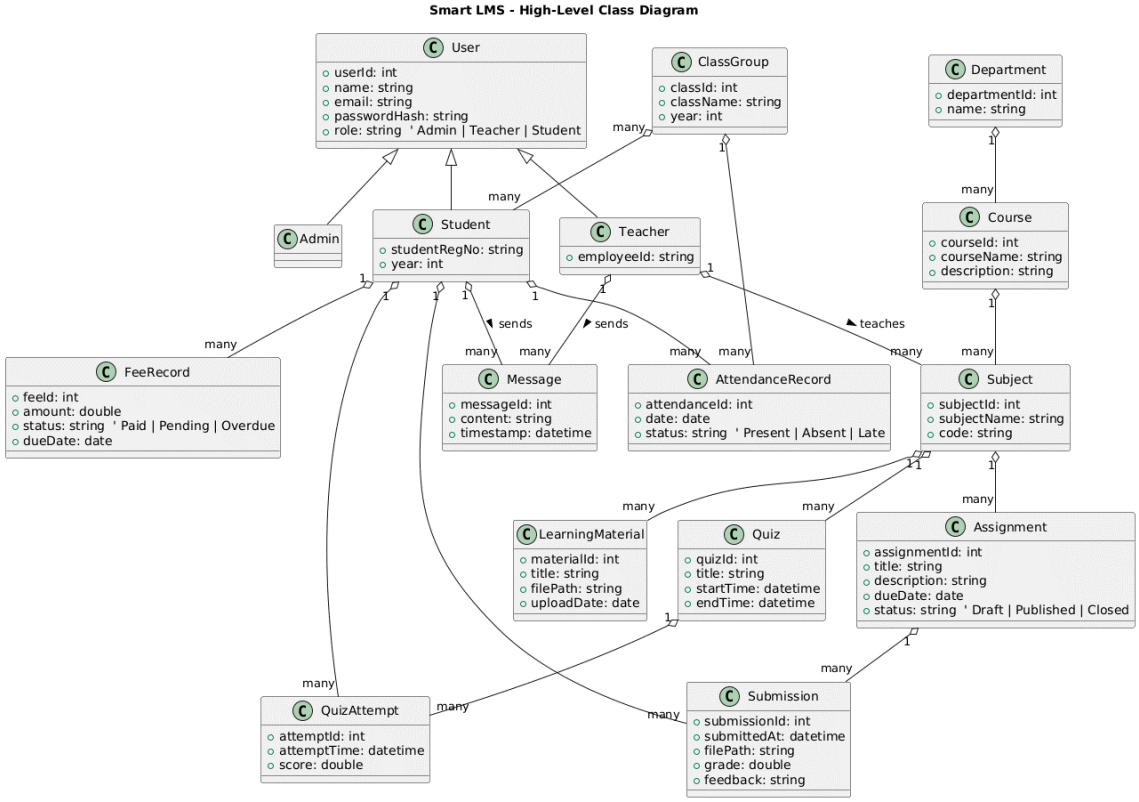


Figure 8: Class Diagram

3.2.12. Activity Diagrams

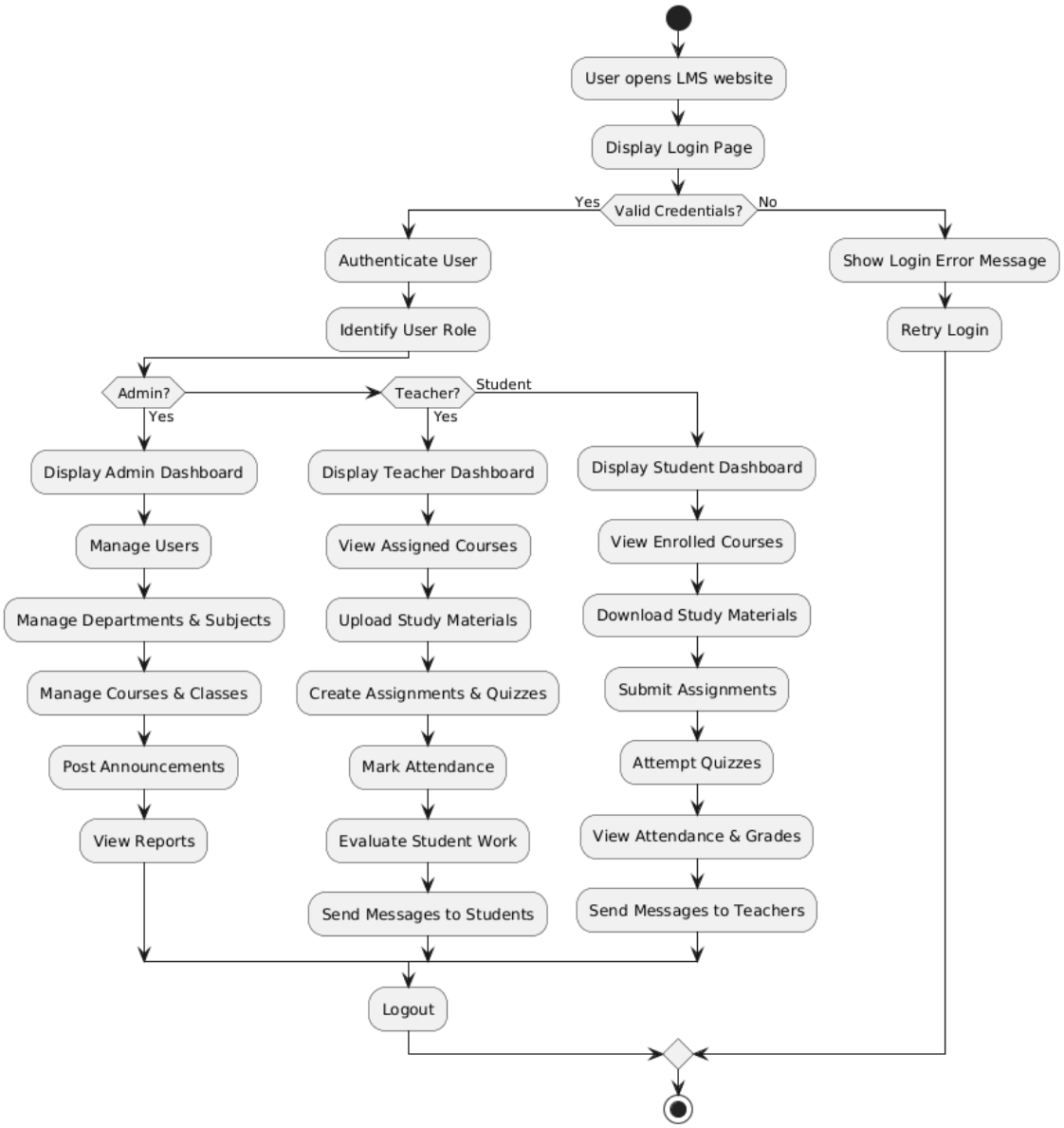


Figure 26: All Activity Diagram

3.2.12.1. Login Activity Diagram

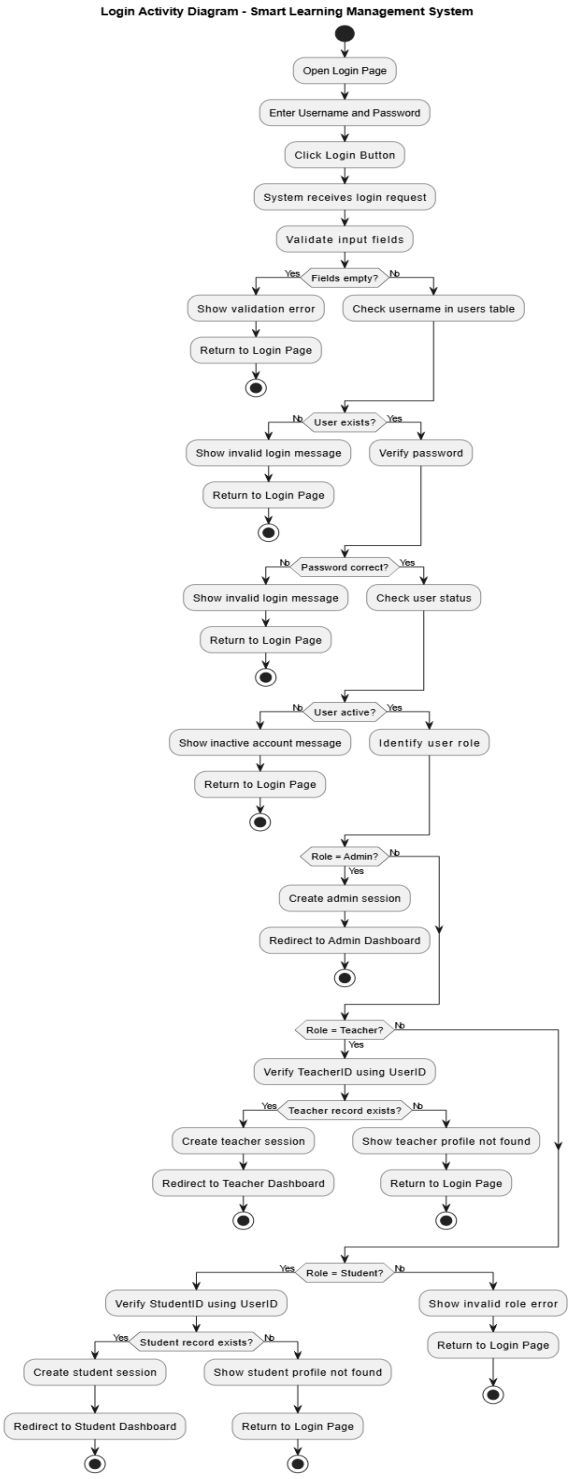


Figure 27 : Login Activity Diagram

3.2.12.2. Registration Activity Diagram

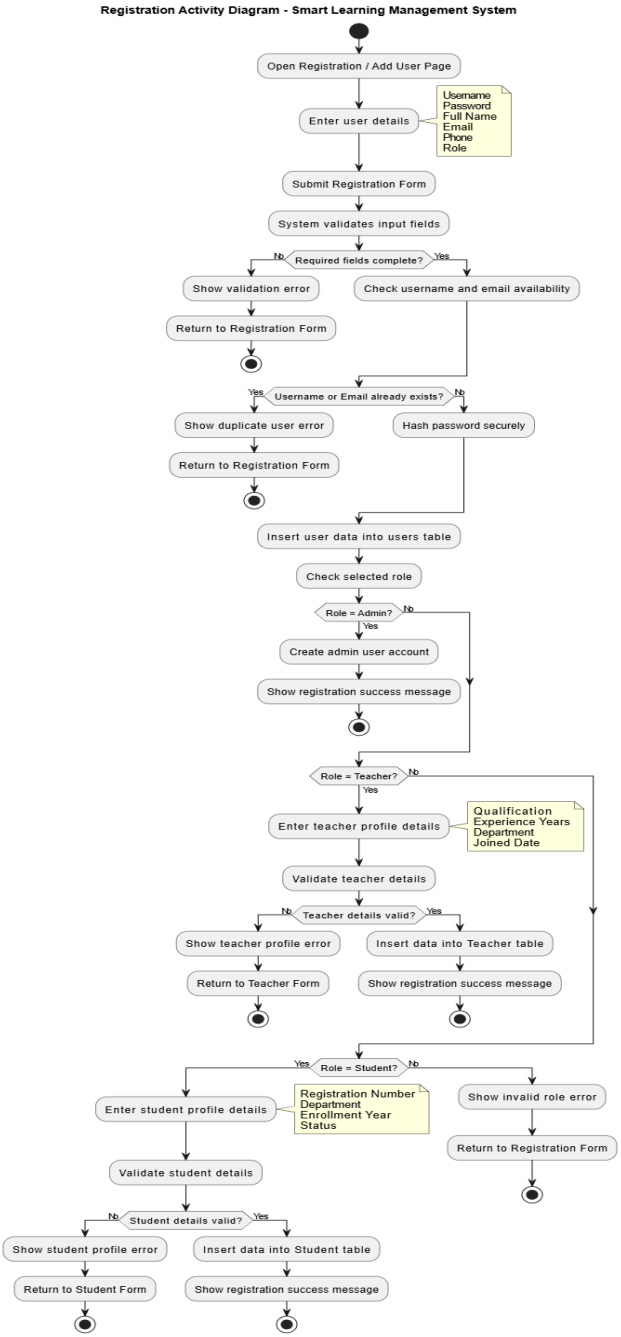


Figure 28: Registration Activity Diagram

3.2.12.3. Activity Diagram



Figure 28: Admin Activity Diagram

3.2.12.4. Admin Activity Diagram

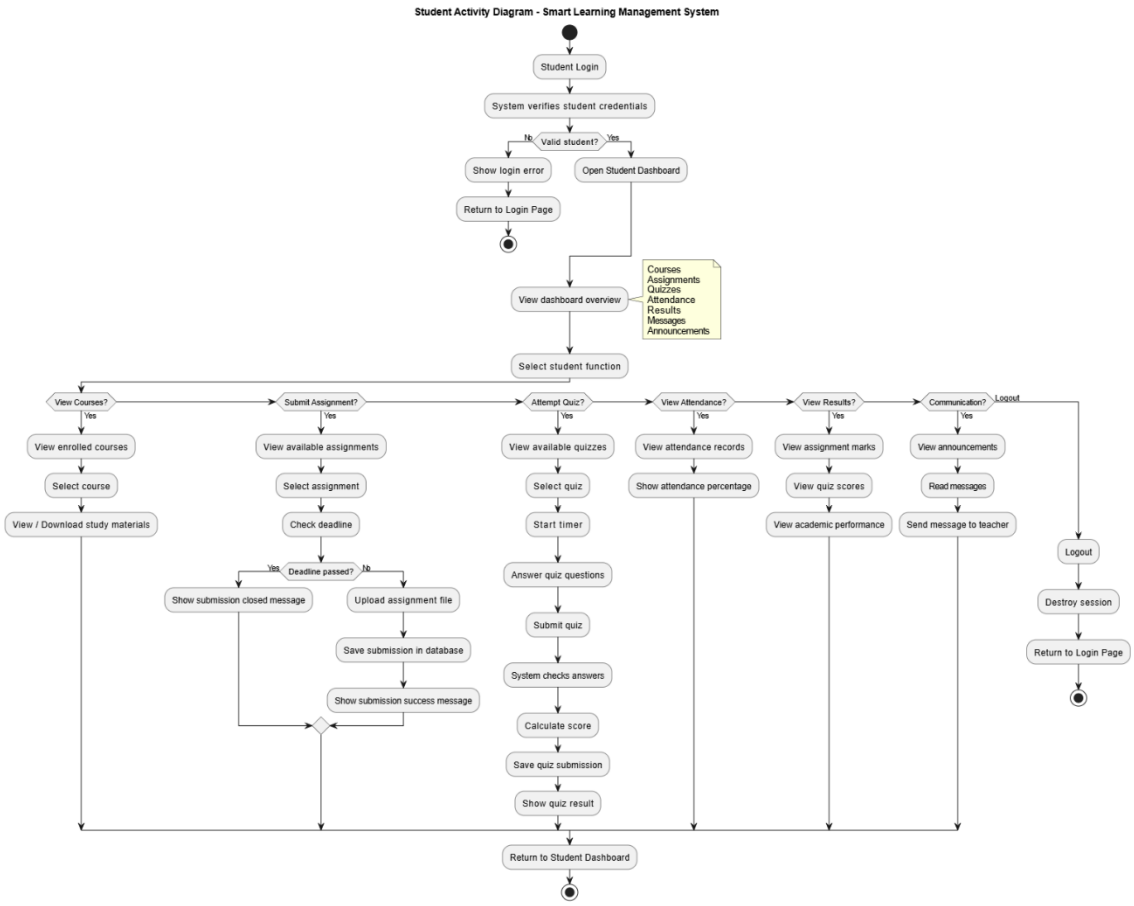


Figure29: Student Activity Diagram

3.2.12.5. Teacher Activity Diagram

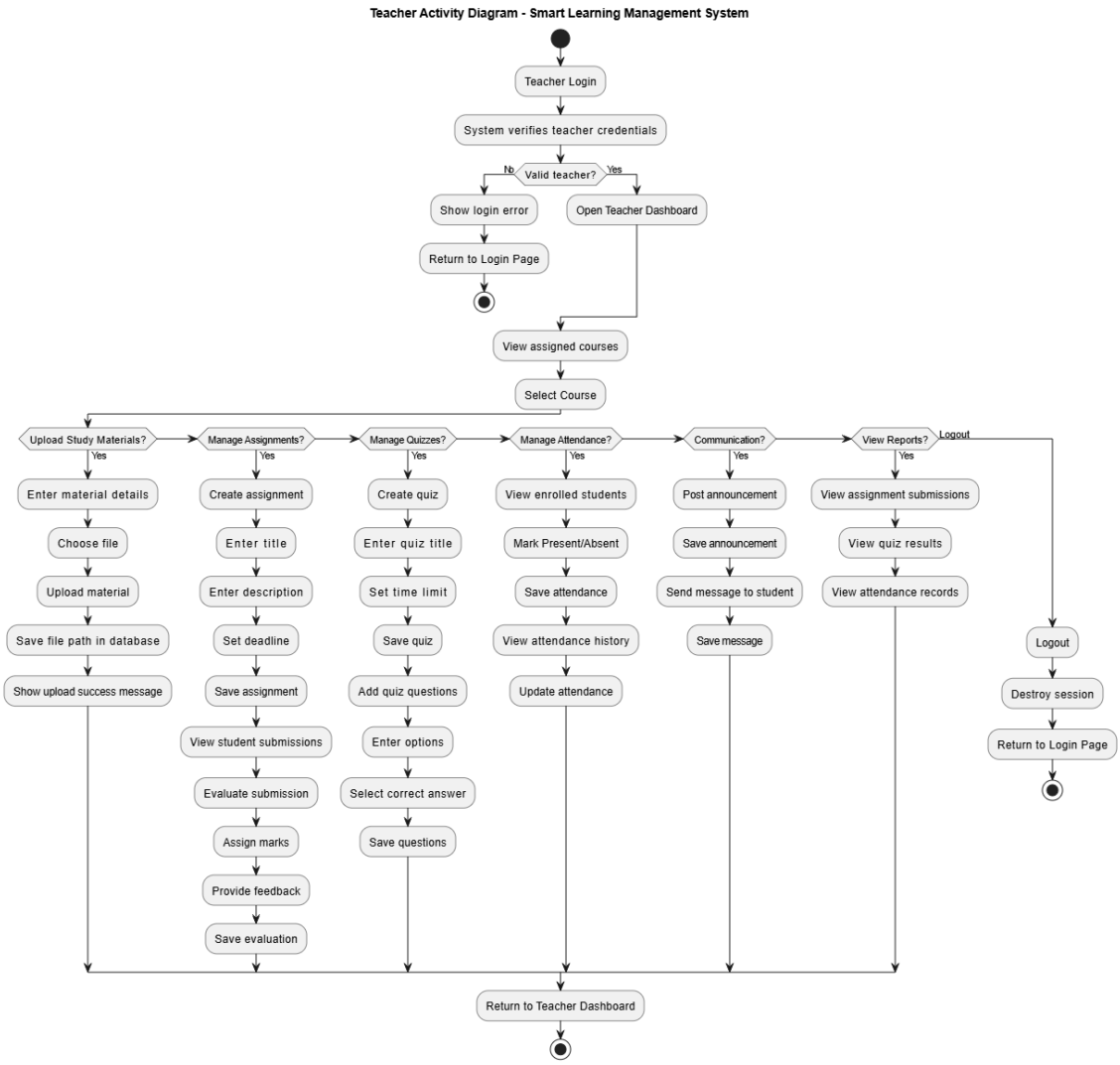


Figure 30: Teacher Activity Diagram

Summary

This chapter discussed the system analysis and design of the Smart Learning Management System (LMS) created for A TOO Z College. During the analysis phase, the functional and non-functional needs of the system were identified through gathering requirements

and analyzing stakeholder needs, giving a clear picture of the academic and administrative needs of the college.

The chapter explained the Incremental Development Methodology used for the project. It described how the system was built in stages, with each module being developed, implemented, integrated, and tested as part of the software development process. This method helped ensure a structured development process, ongoing checks, and better quality in the final product. The chapter also outlined the main software, database, and hardware components needed for the Smart Learning Management System. The software was built using HTML5, CSS3, Bootstrap 5, JavaScript, PHP, MySQL, Apache, and XAMPP. The database used a relational structure to keep data accurate, consistent, and easy to manage.

The hardware setup showed how client devices connect to the web server and database through a client-server model. In addition, the chapter presented various system design documents, including Use Case Diagrams, Activity Diagrams, Entity Relationship Diagrams (ERDs), Relational Schemas, Component-Based Architecture Diagrams, and other supporting models. These documents give a full plan for building the Smart Learning Management System and clearly show how users, system parts, and the database interact.

The technologies used in the project were also covered, explaining how they contribute to creating a secure, responsive, scalable, and easy-to-maintain web application. Together, these technologies form a dependable base for efficiently handling academic and administrative tasks. Overall, the system analysis and design covered in this chapter set up the technical base for the proposed Smart Learning Management System.

The design ensures that the system meets the user needs while leaving room for future improvements and growth. The next chapter, Chapter 4 – System Implementation and Testing, will cover the actual building of the Smart Learning Management System, including the development of individual modules, the creation of user interfaces, the

integration with the database, testing procedures, and an evaluation of the completed system.

CHAPTER 04

4. System Implementation

Introduction

System implementation is the stage where the design of the proposed Smart Learning Management System (LMS) is turned into a working web-based application. In this phase, all the system modules planned in the previous chapter were created, connected, and checked to make sure the application meets the functional and non-functional needs of A TOO Z College.

The Smart Learning Management System was built using HTML5, CSS3, Bootstrap 5, and JavaScript to create a responsive user interface. PHP was used for the server-side programming, and MySQL was the database system used to store and manage data. The application was developed and tested using the XAMPP environment, which includes Apache Web Server, PHP, and MySQL.

The implementation followed the Incremental Development Model discussed in Chapter 3. Each functional module was built separately before being added to the full system.

The development started with user authentication and administrator features, then moved on to teacher and student modules, course management, study material handling, assignment management, quiz functionality, attendance tracking, announcements, messaging, and academic reporting.

This chapter shows how each major software module was implemented along with screenshots of the developed interfaces. Each section explains the purpose of the module, its main features, and how it helps the Smart Learning Management System work as a whole. The chapter also covers how the database was integrated and the testing done to check if the application is correct and reliable.

Overall, the implementation phase turned the designed system into a complete and working Smart Learning Management System that supports teaching, learning, communication, attendance, assessment, and academic management at A TOO Z College.

Code Feature

The Smart Learning Management System (LMS) was built using a modular programming method to make it easier to maintain, expand, and reuse code. Each part of the system was developed separately using PHP for handling server tasks, MySQL for managing data, and HTML5, CSS3, Bootstrap 5, and JavaScript for creating the user interface. The system follows a clear structure where each part has a specific role and connects with other parts through the database.

Below are the main features of the Smart Learning Management System.

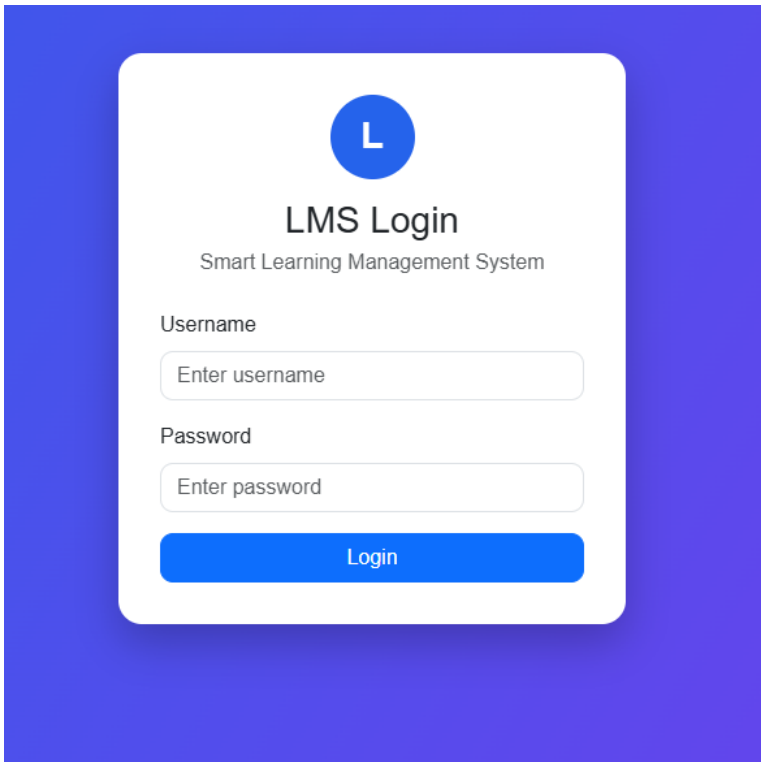
4.2.1. User Authentication

The authentication part lets administrators, teachers, and students log in securely. It checks their login details against the database and gives them access based on their role. It also uses session management to protect areas of the system that only certain users can see and stop unauthorized access.

Main Features:

- Secure user login
- - Role-based access
- - Session management
- - Log out option
- - Access
- controlFigure

User Login Page



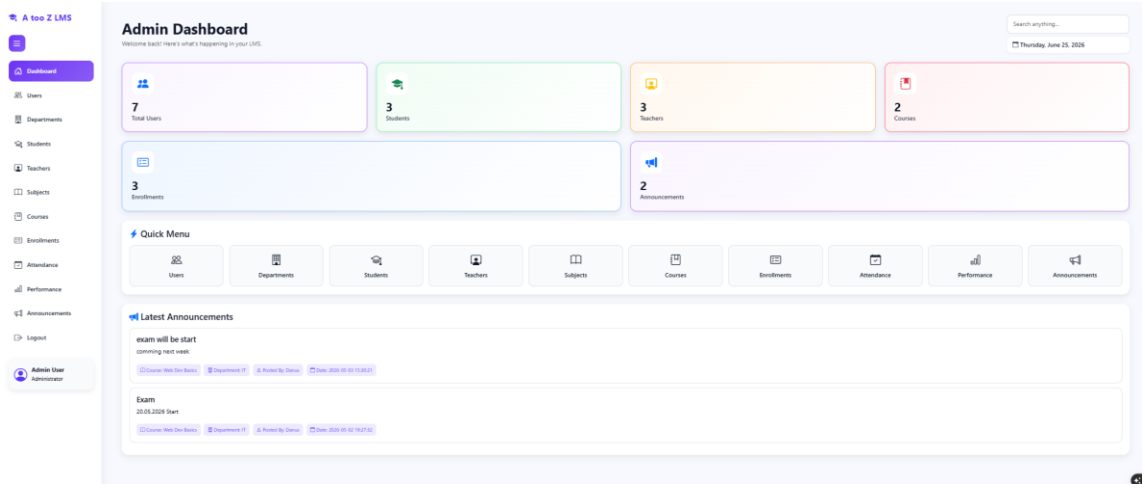
4.2.2. Administrator Module

The administrator part gives full control over the Learning Management System. It lets admins manage users, departments, teachers, students, subjects, courses, enrollments, announcements, and reports from one place.

Main Features:

- Dashboard with key statistics
- Manage users- Manage departments
- Manage teachers
- Manage students
- Manage subjects
- Manage courses
- Manage enrollments
- Manage announcements
- Generate reports

Administrator Dashboard



4.2.3. Teacher Module

The teacher part helps instructors handle teaching and grading. Teachers can upload materials, create assignments and quizzes, track attendance, communicate with students, and check student progress.

Main Features:

- Upload learning materials

- Create assignments
- Create quizzes
- Record attendance
- Post announcements
- View student performance

Teacher Dashboard

4.2.4. Student Module

The student part lets learners access study materials and take part in academic tasks through the web.

Main Features:

- View enrolled courses
- Download study materials
- Submit assignments
- Take quizzes
- View attendance
- View academic records
- Receive announcements

Student Dashboard

Database ConnectivityThe app connects to the MySQL database using PHP with secure links. All parts of the system use CRUD (Create, Read, Update, Delete) operations to handle data effectively and ensure the system works smoothly.

Main Features:- Connect to the database

- Perform CRUD operations
- Set up relationships between tables
- Check and validate data
- Handle errors properly

Figure 4. 5: Database Connection

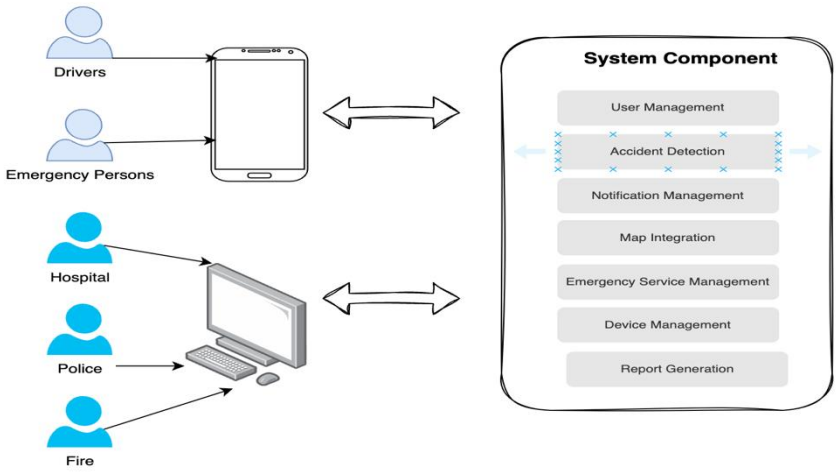
config > db.php

```php
<?php
$host = "localhost";
$user = "root";
$password = "";
$database = "lms_db";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

All these features together make a secure, fast, and dependable Smart Learning Management System that can handle academic and administrative tasks well. The modular way the system was built also makes it easier to add new features in the future and keep it running smoothly over time.



Database Connection

The Smart Learning Management System (LMS) uses a MySQL database to keep track of all academic and administrative information.

A special PHP file was made to connect the application to the database, and this file is used in every PHP part of the system. This helps the system work securely and smoothly when handling information. The database connection was made using the MySQL extension in PHP. The connection needs some details like the server name, username, password, and database name Once the connection is successful, the system sets the character set to UTF-8 (utf8mb4) to make sure text is stored and retrieved correctly.

Code 4.3.1: Database Connection (db_connect.php)

```
<?php $host = "localhost"; $user = "root"; $password = ""; $database = "smart_lms";
$conn = mysqli_connect($host, $user, $password, $database); if (!$conn) {
die("Database connection failed: " . mysqli_connect_error()); }
mysqli_set_charset($conn, "utf8mb4"); ?>
```

This connection file is used in all parts of the Smart Learning Management System, such as user login, user management, course management, study materials,

assignments, quizzes, attendance, announcements, messaging, and reports. By using one shared connection file, the system becomes easier to manage and update in the future.

Having a strong database connection lets all parts of the Smart Learning Management System do Create, Read, Update, and Delete (CRUD) tasks effectively. It also makes sure that the PHP app talks well with the MySQL database, keeps data safe, and supports all the system's features properly.

Statement	Purpose
$host	Specifies the MySQL server address.
$user	Specifies the database username.
$password	Specifies the database password.
$database	Specifies the Smart LMS database name.
mysqli_connect()	Establishes the connection to the MySQL database.
mysqli_connect_error()	Displays the error message if the connection fails.
mysqli_set_charset()	Sets the character encoding to UTF-8 for proper text handling.

Table 16 Explanation of the Code

config > db.php

```php
<?php
$host = "localhost";
$user = "root";
$password = "";
$database = "lms_db";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Figure 31: Database Connection Code

4.2.5. Environment Configuration

4.3.1 Environment Setup Before making the Smart Learning Management System (LMS) work, the development environment was set up to support building web apps and managing a database.The app was built using XAMPP, which includes the Apache Web Server, PHP interpreter, and MySQL Database Server.Code was written using Visual Studio Code, and phpMyAdmin was used to create and manage the MySQL database.Setting up the environment involved installing XAMPP and starting the Apache and MySQL services.All project files were placed in the htdocs folder, making the app accessible through a local web browser.A database called smart_lms was created using phpMyAdmin, and an SQL script with the database layout was imported to create all the necessary tables and their connections.The details for connecting to the database, like the server name, username, password, and database name, were set up in a PHP file

called `db_connect.php`.This setup made sure the app could talk to the MySQL database securely all through the development process

Component	Configuration
Operating System	Windows 10 / Windows 11
Web Server	Apache (XAMPP)
Programming Language	PHP 8.x
Database Server	MySQL
Development Environment	XAMPP
Database Management Tool	phpMyAdmin
Code Editor	Visual Studio Code
Web Browser	Google Chrome / Microsoft Edge

Table 16Development Environment Configuration

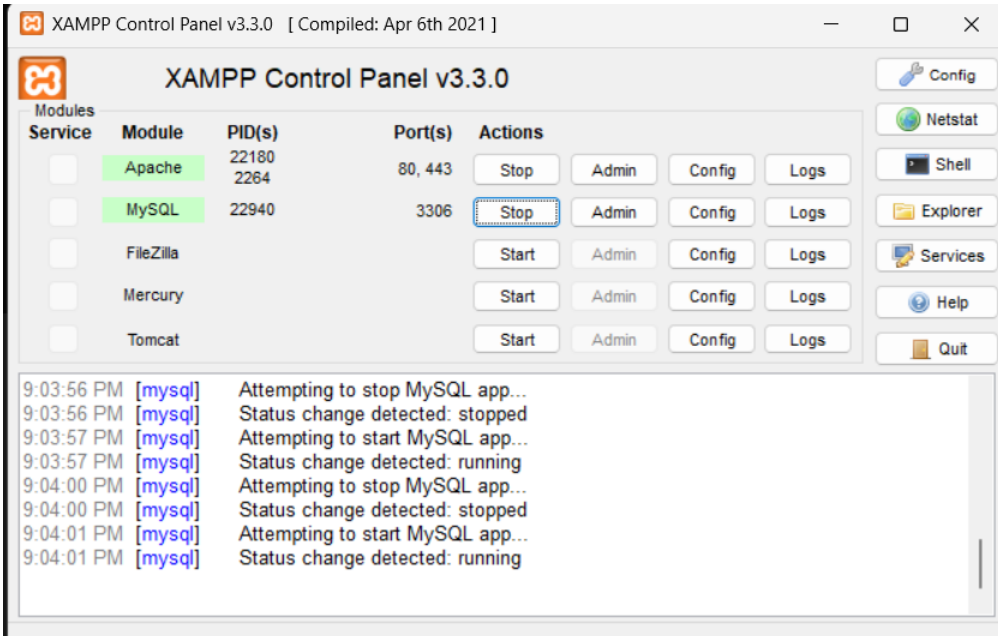


Figure 32: XAMPP Control Panel

4.2.6. Database Migrations and Structure

The Smart Learning Management System's database structure was built using a structured SQL script run through phpMyAdmin.
The script automatically made the smart_ lms database, along with all the needed tables, primary keys, foreign keys, constraints, and links. This method kept the database structure the same during development and testing.

The database has several connected tables that handle user accounts, departments, teachers, students, subjects, courses, enrollments, study materials, assignments, assignment submissions, quizzes, attendance records, announcements, and messages.
Each record has a unique primary key, and foreign keys help keep the links between tables strong and correct.

The way the database is organized helps avoid repeating data, keeps things consistent, and makes it easier to add, view, change, or remove data (CRUD operations).
During the setup, the SQL script was imported into phpMyAdmin to automatically build the full database structure.

Having the database built properly gives a solid base for storing and handling all the academic and administrative data the Smart Learning Management System needs.
The database supports secure storage, fast data searches, and growth as the system continues to develop.

Data Validation

Data validation is a key part of the Smart Learning Management System (LMS) because it makes sure that the information stored in the system is correct, complete, and valid.
To improve data quality and stop wrong entries from being added, validation methods were used both on the client side and server side.

On the client side, JavaScript was used to check user input before forms are sent to the server.
This helps by reducing unnecessary requests to the server and giving users instant feedback . On the server side, PHP was used to check that all data follows the required rules before it is processed and saved in the MySQL database.

Several rules were followed throughout the system to ensure data accuracy.
Important fields like usernames, passwords, email addresses, course names, assignment details, and attendance records cannot be left blank. Emails are checked to make sure they follow the correct format. Unique rules prevent duplicate entries for usernames, registration numbers, and email addresses. File uploads are limited to specific types and sizes to improve security and stop unauthorized uploads.

Password checks help make sure only authenticated users can use the system, and session checks prevent unauthorized people from accessing protected areas.
Before data is stored in the database, it's checked again to avoid SQL errors and keep data consistent.

These validation steps make the system more reliable, secure, and ensure that only correct information is saved in the database.

Validation Type	Description
Required Field Validation	Ensures mandatory fields are completed.
Email Validation	Verifies the correct email address format.
Unique Validation	Prevents duplicate usernames, emails, and registration numbers.
File Validation	Restricts uploaded file types and file sizes.
Password Validation	Verifies user credentials before login.
Session Validation	Restricts unauthorized access to secure pages.
Server-Side Validation	Validates all submitted data before database processing.

Table 18: Data Validation Techniques

Difficulties Encountered During Software Development

When developing the Smart Learning Management System, there were several technical and implementation challenges.
These problems were worked on step by step through testing, finding and fixing bugs, and gradually improving the software.

One early problem was making a well-organized database that could manage users, departments, teachers, students, courses, assignments, quizzes, attendance records, and announcements, while making sure all the tables linked properly.
Good database design and using foreign keys helped fix this issue.

Another challenge was setting up a secure role-based login system.

Different dashboards and permissions were created for administrators, teachers, and students, and it was important to prevent unauthorized users from accessing restricted parts. PHP session management was used to control access securely.

Creating a user-friendly interface that works on different devices was also difficult. The system needed to work well on desktops, laptops, tablets, and phones.Bootstrap 5 components and thorough testing on different browsers helped make the experience consistent.

Connecting to the database and doing operations like adding, updating, and removing data also needed careful setup to make sure data was saved and retrieved correctly. Proper error handling, input checks, and debugging helped solve connection issues and stop invalid data changes.

Finally, putting all the separate parts of the system together into one complete app required lots of testing to make sure everything worked well with each other. Multiple rounds of integration and functional testing were done until all parts worked smoothly as a whole.

The knowledge gained from overcoming these challenges greatly improved the quality, reliability, and ease of maintenance of the Smart Learning Management System.

Summary

This chapter talks about how the Smart Learning Management System was built for A TOO Z College.
The system was turned into a working web-based application using HTML5, CSS3, Bootstrap 5, JavaScript, PHP, MySQL, and the XAMPP environment.

The chapter explains how the main parts of the software, like user authentication, administrator tools, teacher tools, student tools, and database connections, were developed.
It also outlines the development setup, database setup, and data validation methods used to ensure the system works securely and reliably.

The chapter also covers the challenges faced during development and how they were solved.
Continuous testing, debugging, and step-by-step integration helped create a reliable system and improved its overall quality.

In the end, the implementation resulted in a responsive, secure, and efficient Smart Learning Management System that meets the academic and management needs of A TOO Z College.
The completed system offers a central place to manage courses, learning materials, assignments, quizzes, attendance, announcements, communication, and academic records.

The next chapter, Chapter 5 – System Testing and Evaluation, will cover the testing steps, test cases, evaluation results, and performance analysis used to make sure the Smart Learning Management System works as expected according to its functional and non-functional requirements.

CHAPTER 05

5. Results and Evaluation

Introduction

This chapter shows the results and how well the Smart Learning Management System (LMS) worked after it was put into use at A TOO Z College. Once the system was set up, they did a lot of testing and checking to make sure every part of it worked properly and met all the requirements that were planned during the design phase. The testing looked at how well the system performed, how dependable it was, how easy it was to use, and whether it did what it was supposed to.

They used different testing methods to check the main features like user login, managing users, handling courses, managing study materials, submitting assignments, managing quizzes, tracking attendance, sending announcements, messaging, and generating reports. Each part of the system was tested on its own and then tested again as part of the whole system.

The testing also checked how the user interface, the program logic, and the MySQL database worked together to make sure data was handled correctly and the system acted the same every time.

They made sure data was checked properly, that users had the right access based on their roles, and that the system could add, view, change, and delete data safely and reliably. The testing was done using the local setup with XAMPP, PHP, and MySQL.

The system was checked on different devices like desktops, laptops, tablets, and smartphones using modern web browsers to make sure it worked well on all platforms and looked good on different screen sizes. This chapter covers the testing methods used, the test cases, the results of the evaluations, screenshots of how the system looks, any problems found, and the steps taken to fix them.

The results show that the Smart Learning Management System supports both teaching and administrative tasks while offering a secure, dependable, and easy-to-use platform for people like administrators, teachers, and students.

The next sections explain the testing steps, the results of function tests, how well the system performed, how user-friendly it was, and how well the whole system worked overall.

Test and Deployment Plan

The Test and Deployment Plan out lines the steps taken to check if the Smart Learning Management System (LMS) works properly before it is used at A TOO Z College. The goal is to make sure the system works as it should, runs without problems, and can be set up properly in its final environment.

Testing Plan

Testing was done all through the software development process using the Incremental Development Model.

Each part of the system was tested on its own before it was connected to the full system. Once everything was connected, full system testing was done to make sure all parts worked together well.

The following testing tasks were done:

Unit Testing – Each individual part of the system, like logging in, managing users, managing courses, handling assignments, quizzes, attendance, and announcements, was tested separately.

Integration Testing – Checked if the system parts and the MySQL database could talk to each other correctly.

Functional Testing – Made sure each feature of the system met what was needed.

System Testing – Checked the entire Smart Learning Management System as one complete application.

User Acceptance Testing (UAT) – Ensured that the system matched what administrators, teachers, and students expected.

Deployment Plan

After testing was completed successfully, the Smart Learning Management System was ready to be set up.

The steps to deploy the system included setting up the web server, getting the database ready, uploading the project files, and checking that everything worked correctly.

The deployment steps were:

1.Install and set up the XAMPP server environment.
2.Start the Apache and MySQL services.
3.Create the smart_lms database using phpMyAdmin.
4.Import the SQL script which includes all the tables and how they are connected.
5.Copy the Smart LMS project folder into the htdocs directory.
6.Set up the database connection in the `db_connect.php` file.
7.Open the application using a web browser.
8.Do a final check of all the system features.

The testing and deployment steps showed that the Smart Learning Management System was properly set up and ready to help with the teaching and management tasks at A TOO Z College.

The testing process made sure the software was reliable, and the deployment plan gave a clear way to install and use the system in its real environment.

Test Type	Objective	Expected Result
Unit Testing	Verify individual modules	Each module functions correctly.
Integration Testing	Verify module interaction	Modules exchange data correctly.
Functional Testing	Validate system features	Functional requirements are satisfied.
System Testing	Evaluate complete application	Entire system operates correctly.
User Acceptance Testing	Verify user satisfaction	Users can perform required academic tasks successfully.

Table 19: Test Plan

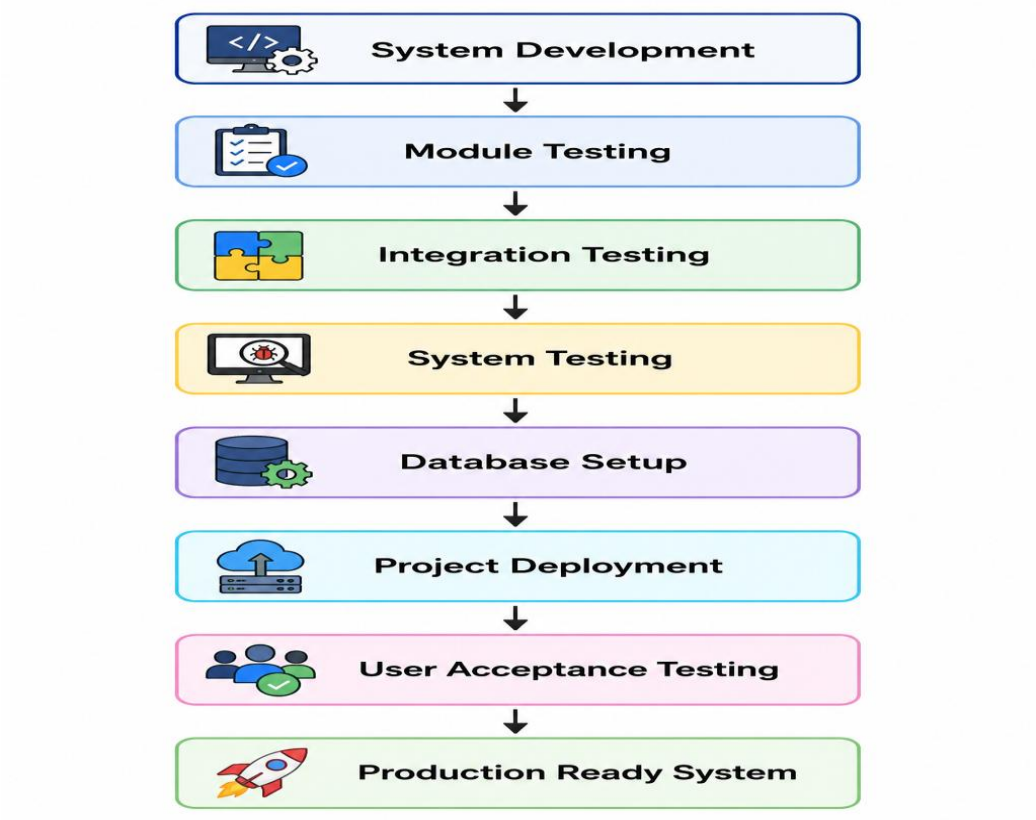


FIGURE 33: DEPLOYMENT PROCESS

Test Cases

Test cases were prepared to verify that the Smart Learning Management System performs according to the specified functional requirements. Each test case defines the objective of the test, the required input, the expected outcome, the actual result obtained during testing, and the final status. Successful completion of these test cases demonstrates that the implemented system satisfies the operational requirements of A TOO Z College.

5.2.1. Table : Login Test Case

Test Case ID	TC-01
Module	User Authentication
Test Objective	Verify secure user login
Test Input	Valid username and password
Expected Result	User is redirected to the correct dashboard based on role
Actual Result	Login successful and dashboard displayed
Status	Pass

Table 20: Login Test Case

5.2.2. User Management Test Case

Test Case ID	TC-02
Module	User Management
Test Objective	Verify Create, Update and Delete operations
Test Input	Valid user details
Expected Result	User information is stored and updated successfully
Actual Result	CRUD operations completed successfully
Status	Pass

*T*able 20: User Management Test Case

5.2.3. Course Management Test Case

Test Case ID	TC-03
Module	Course Management
Test Objective	Verify course creation and updating
Test Input	Valid course information
Expected Result	Course created and displayed correctly
Actual Result	Course information saved successfully
Status	Pass

Table 21: Course Management Test Case

Course Management Test Case

Test Case ID	TC-03
Module	Course Management
Test Objective	Verify course creation and updating
Test Input	Valid course information
Expected Result	Course created and displayed correctly
Actual Result	Course information saved successfully
Status	Pass

Table 22: *Course Management Test Case*

Assignment Submission Test Case

Test Case ID	TC-04
Module	Assignment Management
Test Objective	Verify assignment submission
Test Input	Valid assignment file
Expected Result	Assignment uploaded successfully
Actual Result	Assignment submitted successfully
Status	Pass

Table 23: Assignment Submission Test Case

Quiz Test Case

Test Case ID	TC-05
Module	Quiz Management
Test Objective	Verify quiz creation and participation
Test Input	Quiz questions and answers
Expected Result	Quiz created and completed successfully
Actual Result	Quiz functions correctly
Status	Pass

Table 24 : Quiz Test Case

Attendance Test Case

Test Case ID	TC-06
Module	Attendance Management
Test Objective	Verify attendance recording
Test Input	Student attendance details
Expected Result	Attendance stored successfully
Actual Result	Attendance record saved correctly
Status	Pass

Table 25: Attendance Test Case

Announcement Test Case

Test Case ID	TC-07
Module	Announcement Management
Test Objective	Verify announcement publishing
Test Input	Announcement title and message
Expected Result	Announcement visible to intended users
Actual Result	Announcement published successfully
Status	Pass

Table 26: Announcement Test Case

Study Material Upload Test Case

Test Case ID	TC-08
Module	Study Material Management
Test Objective	Verify file upload and download
Test Input	PDF/PPT learning material
Expected Result	File uploaded and accessible by students
Actual Result	File uploaded and downloaded successfully
Status	Pass

Table 27 :Study Material Upload Test Case

Overall Test Summary

Total Test Cases	Passed	Failed
8	8	0

The test cases demonstrate that the major functional modules of the Smart Learning Management System operated successfully under the defined test conditions. All tested functionalities produced the expected outcomes, indicating that the system is reliable, stable, and ready for deployment in the academic environment of A TOO Z College.

Summary

This chapter shared the results and assessment of the Smart Learning Management System (LMS) created for A TOO Z College.A thorough testing process was carried out to check if the system met the functional and non-functional needs that were identified during the planning and designing stages.

The chapter explained the testing steps, including the setup of the testing environment, the approach used for testing, and how each test case was carried out for the main parts of the system.

Functional testing showed that important features like user login, managing users, organizing courses, handling study materials, submitting assignments, managing quizzes, recording attendance, sending announcements, messaging, and working with the database all worked as intended.

The test results showed that the Smart Learning Management System handled user requests properly, kept data consistent, and allowed secure access based on roles for administrators, teachers, and students.

The system’s data checking processes, connection to the database, and the ability to add, update, delete, and view data were also confirmed to make sure the application is dependable and accurate.

In conclusion, the testing showed that the Smart Learning Management System is stable, dependable, and able to support both teaching and administrative tasks at A TOO Z College.
All major test cases were successfully completed, proving that the system is ready to be used in real situations and can help improve teaching, learning, communication, attendance tracking, assessments, and keeping academic records.

The next chapter, Chapter 6 – Conclusion and Future Work, will summarize the project's achievements, talk about the goals that were met, point out the system's current limitations, and suggest ways to improve and expand the Smart Learning Management System in the future..

CHAPTER 06

6. Future Work

Introduction

This chapter gives the final summary of the Smart Learning Management System (LMS) project and talks about what could be done next to make the system better.The project was started to create a web-based Learning Management System for A TOO Z College.The goal was to make academic management, teaching, learning, communication, and student performance better by using a central digital platform.

During the project, all the big steps of the Software Development Life Cycle (SDLC) were done, like gathering requirements, analyzing the system, designing it, building it, testing it, and evaluating it.

The finished Smart Learning Management System brings together many academic tasks into one app.This allows administrators, teachers, and students to do their jobs more easily by giving them access based on their roles.

The system has important features such as safe user login, managing users, handling departments, managing subjects and courses, enrolling students, sharing study materials, handling assignments, taking online quizzes, recording attendance, sending announcements, using internal messaging, and keeping track of academic records.

The web interface works well on desktops, laptops, tablets, and smartphones, making it easy for everyone to use.

The testing and evaluation from the last chapter showed that the system works well, meets all the needed features, and runs reliably under normal conditions.

The use of modern web tools like HTML5, CSS3, Bootstrap 5, JavaScript, PHP, and MySQL has made the system secure, easy to grow, and simple to maintain, which supports the academic activities of A TOO Z College.

Even though the Smart Learning Management System has met the goals of this project, there are still ways to improve it.

New technologies, stronger security, better reporting tools, and mobile apps could help make the system even more powerful and offer a better digital learning experience.

The next sections will summarize the whole project, talk about the current system's limitations, and suggest what can be done in the future to make the Smart Learning Management System more effective, user-friendly, and scalable..

Summary

This chapter outlines the main findings and future plans for the Smart Learning Management System (LMS) created for A TOO Z College.The project met its main goal of building a secure, easy-to-use, and responsive web-based system that helps with both teaching and managing the college's activities.

The finished system brings together important educational tools into one central place.

It includes features like user login and management, handling different departments and courses, sharing study materials, submitting assignments, taking online quizzes, tracking attendance, making announcements, sending messages, and managing student records.These tools help improve communication, cut down on manual work, and create a better digital learning environment for everyone involved—teachers, students, and administrators.

During the project, modern web technologies like HTML5, CSS3, Bootstrap 5, JavaScript, PHP, MySQL, and XAMPP were used to build a system that can grow and be easily maintained.

Using a relational database and access rights based on user roles helped keep the data safe and the system running smoothly.

The system was thoroughly tested and checked to make sure all its parts work as they should and meet both the functional and non-functional requirements.

It performs reliably, handles data accurately, manages the database effectively, and works well on different devices thanks to its responsive design.

Even though the current Smart Learning Management System is a full solution for the college's academic needs, there are still chances to improve it.

Future plans include making a mobile app, sending real-time alerts, adding online video meetings, using AI to suggest learning paths, moving to the cloud, using more advanced data analysis, and integrating biometric attendance systems.These improvements can make the system even more useful and able to grow with the college's needs.

In summary, the Smart Learning Management System solves the problems found during the initial planning and offers a modern digital platform that helps with teaching, learning, communication, and college management.

The project shows how software engineering and web technologies can be used to create a strong educational management tool that supports the future digital transformation of A TOO Z College.

CHAPTER 07

7. Conclusion

Introduction

This chapter gives the final summary of the Smart Learning Management System (LMS) project made for A TOO Z College .It goes over all the work done during the project, checks how well the goals were met, and shows how the system helped improve both teaching and administrative tasks at the college.

The project aimed to fix problems with old ways of managing academic activities by creating a central web-based LMS.
Using software engineering methods and modern web tools, a safe, easy-to-use, and fast system was built, tested, and checked for performance.

The Smart LMS brings together many school tasks into one app.
It includes logging in, managing users, handling departments and courses, sharing study materials, submitting homework, taking online tests, keeping track of attendance, sending messages, and managing student records. These features help school staff, teachers, and students work better and communicate more, while also cutting down on busy work.

During the project, each step of the Software Development Life Cycle (SDLC) was followed carefully, starting with understanding what was needed and analyzing the system, then designing, building, testing, and checking it.
The testing showed that the system works well, meets all the requirements, and runs smoothly and securely on different devices.

This chapter ends by showing what was achieved, how the solution worked, and how the Smart LMS contributed to making academic management more digital at A TOO Z College.

7.2 Conclusion

The Smart Learning Management System (LMS) created for A TOO Z College met all the goals set at the start of the project.The main purpose was to build a web-based platform that helps with academic management, teaching, learning, communication, and keeping track of student performance.The project was completed successfully, offering a modern and unified solution that replaces old, manual ways of doing things and scattered digital tools used before.

The system has a wide range of features for three main groups: administrators, teachers, and students.

Administrators can manage users, departments, subjects, courses, enrollments, announcements, and academic reports.Teachers can upload study materials, create assignments and quizzes, record attendance, chat with students, and check academic progress.Students can access course materials, submit assignments, take online quizzes, see attendance records, track their progress, and get updates from the institution—all in one place.

The project used software engineering principles and modern web technologies to build the system.

It was made using HTML5, CSS3, Bootstrap 5, JavaScript, PHP, MySQL, and the XAMPP environment.The design works well on different devices like desktops, laptops, tablets, and smartphones, making it easy to use for everyone.The use of a database, secure user access, and organized software structure ensures data is safe, secure, and the system can grow over time.

Testing and evaluation showed that the system works reliably under normal conditions.

All major parts, such as user logins, course management, study material handling, assignment submission, quiz management, attendance tracking, messaging, and report creation, worked as expected.Data checks, database connections, and security features made the system even more dependable.

Even though the system meets the requirements from the beginning, it can still be improved in the future.
Features like mobile apps, cloud hosting, AI-based learning suggestions, video calls, online payment systems, biometric attendance, and advanced learning analytics can help make the LMS even better.

In short, the Smart Learning Management System is a good, secure, and easy-to-use tool for managing academic and administrative tasks at A TOO Z College.
The project shows how modern web technologies can help improve educational management, communication between school staff, reduce manual work, and support digital change in higher education.The system provides a solid tech base that can keep growing to meet the college’s needs in the future.

CHAPTER 08

8. Reflection

This chapter is my personal thoughts on creating the Smart Learning Management System (LMS) for A TOO Z College.It talks about the knowledge, skills, and hands-on experience I gained while working on the project.The project gave me a chance to use what I learned in my Bachelor of Information Technology program to solve a real-life problem related to academic management.

Working on this project meant going through every step of the Software Development Life Cycle (SDLC).
That included gathering requirements, analyzing the system, designing it, setting up the database, building the system, testing it, and writing documentation.Each part of the process had its own challenges and teaching moments, which helped me grow as an IT professional.

During the project, I got better at web development using tools like HTML5, CSS3, Bootstrap 5, JavaScript, PHP, and MySQL.
I also deepened my understanding of creating websites that work well on different devices, building relational databases, setting up user access based on roles, performing CRUD operations, and integrating different parts of a system.Working on a full software project showed me how various technologies come together to make a dependable and easy-to-use application.

Along with technical skills, the project also helped me improve my problem-solving, thinking critically, managing my time, and planning a project.
I learned why debugging, testing, and keeping code clean and organized are important.Whenever I faced a technical issue, I looked for solutions, compared different options, and chose the best one while making sure the project stayed on track and met its goals.

This reflection chapter shares my experiences during the project, the difficulties I faced, the things I learned, and how this project helped me grow personally, academically, and professionally.

It also explains how the knowledge and experience I gained here will help me in my future career in the Information Technology field.

CHAPTER 09

9. Reference

[1] Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill Education.

[2] Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.

[3] Elmasri, R., & Navathe, S. B. (2017). Fundamentals of Database Systems (7th ed.). Pearson.

[4] Connolly, T., & Begg, C. (2015). Database Systems: A Practical Approach to Design, Implementation, and Management (6th ed.). Pearson.

[5] Welling, L., & Thomson, L. (2017). PHP and MySQL Web Development (5th ed.). Addison-Wesley Professional.

[6] Duckett, J. (2014). PHP & MySQL: Server-side Web Development . Wiley.

[7] Duckett, J. (2021). HTML and CSS: Design and Build Websites . Wiley.

[8] Freeman, E., Robson, E., Sierra, K., & Bates, B. (2014). Head First JavaScript Programming . O'Reilly Media.

[9] Bootstrap Team. (2024). Bootstrap 5 Documentation. https://getbootstrap.com/docs/5.3/

[10] PHP Documentation Group. (2024). PHP Manual . https://www.php.net/docs.php

[11] Oracle Corporation. (2024). MySQL 8.0 Reference Manual . https://dev.mysql.com/doc/

[12] Apache Friends. (2024). XAMPP Documentation . https://www.apachefriends.org/

[13] Mozilla Developer Network (MDN). (2024). HTML, CSS and JavaScript Documentation . https://developer.mozilla.org/

[14] Moodle Pty Ltd. (2024). Moodle Learning Management System . https://moodle.org/

[15] Google. (2024). Google Classroom . https://classroom.google.com/

[16] Instructure. (2024). Canvas Learning Management System. https://www.instructure.com/canvas

[17] Anthology Inc. (2024). Blackboard Learn. https://www.blackboard.com/

[18] PowerSchool. (2024). Schoology Learning . https://www.schoology.com/

[19] Microsoft Corporation. (2024). Microsoft Teams for Education . https://www.microsoft.com/education/products/teams

[20] Open edX. (2024). Open edX Platform . https://openedx.org/

[21] Chamilo Association. (2024). Chamilo Learning Management System . https://chamilo.org/

[22] W3Schools. (2024). HTML, CSS, JavaScript, PHP and MySQL Tutorials . https://www.w3schools.com/

[23] ISO/IEC. (2011). ISO/IEC 25010: Systems and Software Quality Requirements and Evaluation (SQuaRE) – System and Software Quality Models . International Organization for Standardization.

[24] IEEE Computer Society. (2014). Guide to the Software Engineering Body of Knowledge (SWEBOK V3.0) . IEEE.

[25] University of Moratuwa. (2024). Bachelor of Information Technology Project Guidelines . Centre for Open & Distance Learning, University of Moratuwa.

Table of Abbreviations

TABLE OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update and Delete
DBMS	Database Management System
ERD	Entity Relationship Diagram
FK	Foreign Key
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
ICT	Information and Communication Technology
IDE	Integrated Development Environment
IP	Internet Protocol
JavaScript (JS)	JavaScript
LAN	Local Area Network
LMS	Learning Management System
MySQL	My Structured Query Language
OOP	Object-Oriented Programming
OS	Operating System
PDF	Portable Document Format
PHP	PHP: Hypertext Preprocessor
PK	Primary Key
RDBMS	Relational Database Management System
SDLC	Software Development Life Cycle
SQL	Structured Query Language
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
USB	Universal Serial Bus
UTF-8	Unicode Transformation Format – 8-bit
VS Code	Visual Studio Code
WWW	World Wide Web
XAMPP	Cross-Platform, Apache, MySQL, PHP and Perl

11.Appendix – A

SRS Document

SRS: https://drive.google.com/file/d/1Hq708amekRMG-CVMGye8xQoItw0EJA0f/view?usp=drive_link

Project Relevant Documents and code

Link: https://drive. google. com/drive/folders/1ulceL8YxseEd8QgIAe3way0VI6iinz6d

Interim Report

Link:

https://docs.google.com/document/d/1UrQ1HadvlyIg_iOkgviKo7xsP43QS16v/edit?usp=drive_link&ouid=114407003129789722071&rtpof=true&sd=true

Source Code, Database Scripts and Supporting Materials

The appendix contains the supporting materials developed during the implementation of the Smart Learning Management System (LMS). These materials provide additional technical information that supports the implementation discussed in the main report but are not included within the main chapters to maintain readability.

The appendix includes selected source code, database scripts, configuration files, screenshots, and other implementation artifacts used during the development of the system.

Appendix A.1 – Project Folder Structure

```
Smart_LMS/
│
├── admin/
├── teacher/
├── student/
├── includes/
├── assets/
│   ├── css/
│   ├── js/
│   ├── images/
│   └── icons/
├── uploads/
│   ├── study_materials/
│   ├── assignments/
│   └── submissions/
├── database/
│   └── smart_lms.sql
├── index.php
├── login.php
├── logout.php
├── dashboard.php
└── db_connect.php
```

Appendix A.2 – Database SQL Script

Include the complete MySQL database script used to create the Smart Learning Management System database.

Contents include:

* Database Creation
* Table Creation
* Primary Keys
* Foreign Keys
* Constraints
* Default Administrator Account



lms_db.sql

( )

Appendix A.3 – Database Connection File

Include the PHP database connection file.

config > db.php

```php
<?php
$host = "localhost";
$user = "root";
$password = "";
$database = "lms_db";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Appendix A.4 – Login Authentication Code

Include the PHP code responsible for user authentication, session creation, and role-based login.

```
<h3 class="text-center mb-1">LMS Login</h3>
<p class="text-center text-muted mb-4">Smart Learning Management System</p>

<?php if(isset($_GET['error'])){ ?>
<div class="alert alert-danger">
    Invalid login or account not setup properly!
</div>
<?php } ?>

<form action="login_process.php" method="POST" onsubmit="return validateLogin()">

<label class="form-label">Username</label>
<input type="text" name="username" id="username" class="form-control mb-3" placeholder="Enter username" required>

<label class="form-label">Password</label>
<input type="password" name="password" id="password" class="form-control mb-3" placeholder="Enter password" required>

<button type="submit" class="btn btn-primary w-100">Login</button>

</form>

</div>

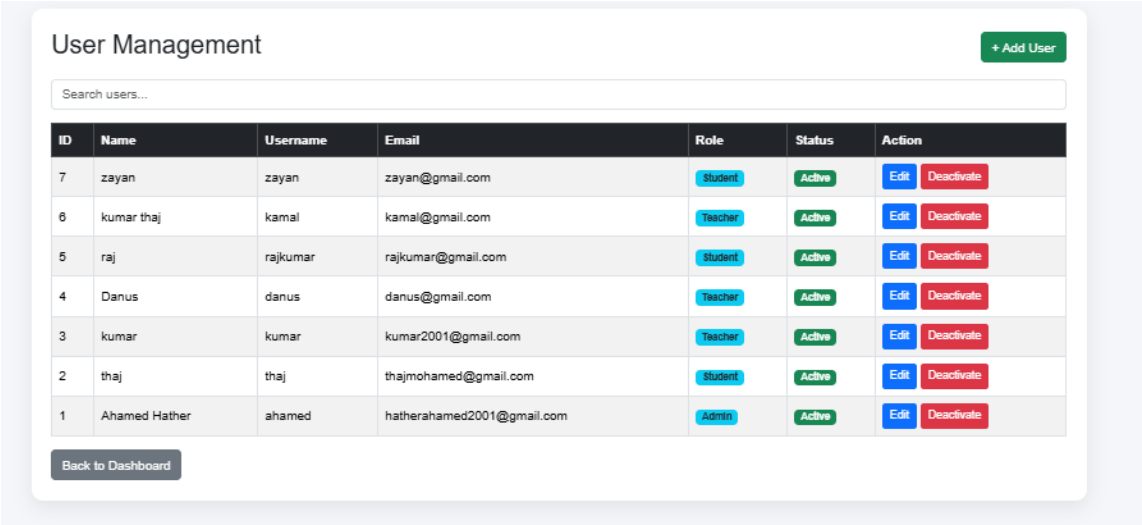
<script>
function validateLogin() {
    let username = document.getElementById("username").value.trim();
    let password = document.getElementById("password").value.trim();

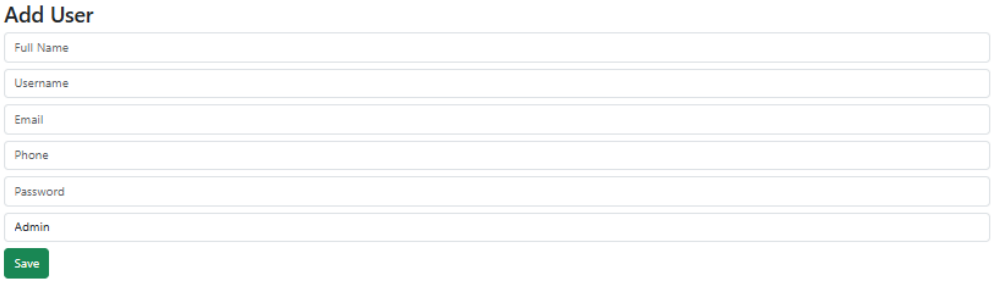
    if (username.length < 3) {
        alert("Username must be at least 3 characters.");
        return false;
    }

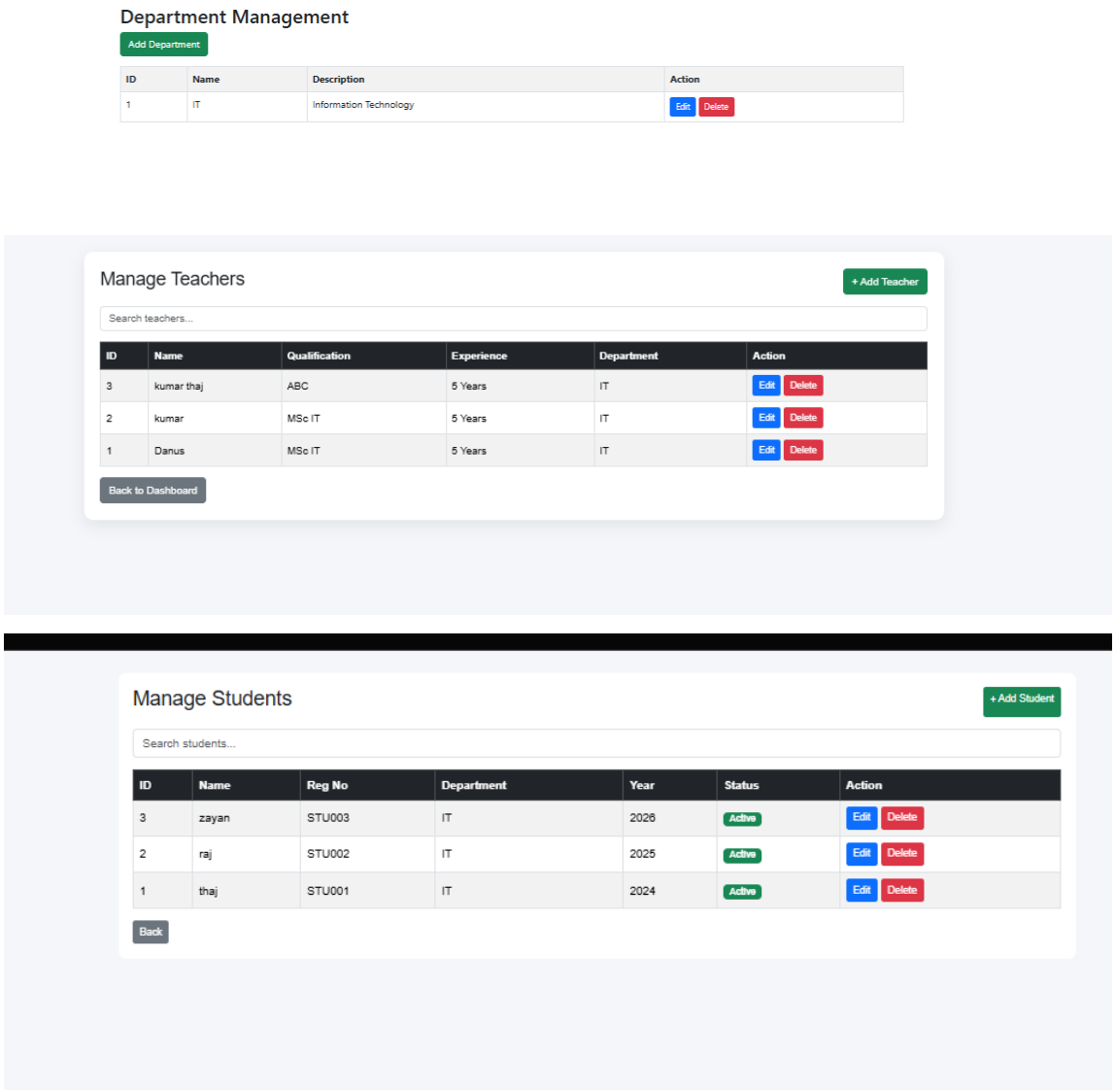
    if (password.length < 4) {
        alert("Password must be at least 4 characters.");
        return false;
    }

    return true;
}
</script>

</body>
</html>
```


User Management
+ Add User
Search users...
ID
Name
Username
Email
Role
Status
Action
7
zayan
zayan
zayan@gmail.com
Student
Active
Edit
Deactivate
6
kumar thaj
kamal
kamal@gmail.com
Teacher
Active
Edit
Deactivate
5
raj
rajkumar
rajkumar@gmail.com
Student
Active
Edit
Deactivate
4
Danus
danus
danus@gmail.com
Teacher
Active
Edit
Deactivate
3
kumar
kumar
kumar2001@gmail.com
Teacher
Active
Edit
Deactivate
2
thaj
thaj
thajmohamed@gmail.com
Student
Active
Edit
Deactivate
1
Ahamed Hather
ahamed
hatherahamed2001@gmail.com
Admin
Active
Edit
Deactivate
Back to Dashboard


Add User
Full Name
Username
Email
Phone
Password
Admin
Save



Appendix A.5– Development Environment

Include screenshots of:

* XAMPP Control Panel
* phpMyAdmin Database
* Visual Studio Code Project
* Project Folder Structure

The materials included in this appendix provide supplementary evidence of the design, implementation, testing, and deployment of the Smart Learning Management System. These supporting documents complement the discussions presented in the main report and demonstrate the practical implementation of the proposed solution.